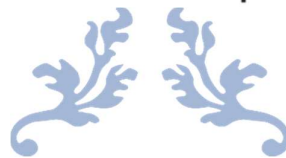




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**Artificial Intelligence–Driven Customer  
Segmentation and Personalized Marketing for  
Handmade Small Businesses**

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Submitted in partial fulfilment of the requirements for the award of the degree  
of

Master of Business Administration (MBA)

By

Student Name: Amit Kumar Srivastava

Enrollment No: O24MBA113874

Under the guidance of

Guide/Mentor Name: Vasanthi Chandran

## **Certificate**

This is to certify that the project titled “*Artificial Intelligence–Driven Customer Segmentation and Personalized Marketing for Handmade Small Businesses*” is a bonafide work carried out by **Amit Kumar Srivastava**, a student of **Master of Business Administration (MBA), 4th Semester**, at **Chandigarh University** during the academic year **2025–2026**, in partial fulfillment of the requirements for the award of the MBA degree.

This project report has been prepared under proper guidance and supervision. To the best of my knowledge and belief, the work presented in this report is original and has not been submitted previously for the award of any degree, diploma, or certificate of this or any other university or institution.

The objective of this project is to develop an **Artificial Intelligence–based customer segmentation model** for handmade small businesses and to improve marketing strategies through personalized customer engagement. The study focuses on analysing customer data such as demographic information, purchasing behaviour, and educational backgrounds to create targeted marketing approaches.

During the course of this project, the student successfully worked on collecting and pre-processing customer data, implementing machine learning algorithms to segment customers based on their education level, interests, and preferences, and designing personalized marketing campaigns for different customer groups.

The project also evaluates the effectiveness of AI-driven marketing strategies on **customer engagement, satisfaction, and loyalty** using analytical tools and relevant performance metrics. The work demonstrates the student’s understanding of modern digital marketing practices and the application of Artificial Intelligence in business decision-making.

I am satisfied with the quality and presentation of this project report and recommend it for submission as a part of the academic requirements for the **MBA program at Chandigarh University**.

**Project Guide:** Vasanthi Chandran

**Designation:** Senior IT Project Manager

**Department:** IT Sector

**Institution:** Chandigarh University

**Signature:**

**Date:** 05/05/2026

## **Declaration**

I, **Amit Kumar Srivastava**, hereby solemnly declare that the project report titled “*Artificial Intelligence–Driven Customer Segmentation and Personalized Marketing for Handmade Small Businesses*” submitted in partial fulfilment of the requirements for the award of the degree of **Master of Business Administration (MBA)** is my original work.

I further declare that:

- This project has been carried out by me during the academic year 2025–26 under the supervision of **Vasanthi Chandran**.
- The work has not been submitted previously to any other university, institution, or examination body for the award of any degree, diploma, or certification.
- All sources of information used in this report have been duly acknowledged and referenced in accordance with academic ethics and plagiarism norms.
- The data presented in this report is authentic to the best of my knowledge and has not been fabricated or manipulated.

Place: Azamgarh, Uttar Pradesh

Date: 05/05/2026

Signature: Amit Kumar Srivastava

Name: Amit Kumar Srivastava

Enrollment No.: O24MBA113874

## **Acknowledgement**

I would like to express my deepest gratitude to my mentor, **Vasanthi Chandran**, Senior Mentor, for his invaluable guidance, encouragement, and expertise throughout the course of this project. His 28 years of diverse leadership experience in marketing, advertising, brand strategy, and strategic partnerships provided me with profound insights that greatly enriched the quality and direction of this research.

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Finally, I am indebted to my family and friends for their unwavering encouragement, patience, and moral support, which sustained me throughout this journey.

This project is the result of collective guidance, cooperation, and inspiration, and I sincerely thank everyone who contributed to its successful completion.

## **Abstract / Executive Summary**

This study investigates the role of Artificial Intelligence (AI)-driven customer segmentation and personalized marketing in strengthening the competitiveness of handmade small businesses in India. The handmade sector, deeply rooted in cultural heritage, faces challenges such as limited resources, fragmented markets, and competition from mass-produced goods.

**Objective:** The primary aim is to develop a machine learning model that segments customers based on educational backgrounds and preferences, and to enhance personalization in marketing strategies for handmade enterprises.

**Methodology:** A mixed-method approach was adopted, combining primary data from surveys of 150 respondents with secondary data from industry reports and academic literature. Statistical techniques such as regression, correlation, and cluster analysis were applied to validate hypotheses and derive actionable insights.

**Key Findings:** Results indicate that AI-driven segmentation provides more accurate and meaningful customer groups compared to traditional demographic methods. Personalized marketing strategies significantly improve customer engagement, loyalty, and repeat purchases. Furthermore, affordable AI tools can be integrated into small enterprises without requiring extensive technical expertise, enabling resource optimization and digital market expansion.

**Conclusion:** The study concludes that AI-driven segmentation and personalized marketing can empower handmade businesses to achieve sustainable growth, preserve cultural heritage, and compete effectively in digital marketplaces. By bridging tradition with technology, small enterprises can enhance customer satisfaction while contributing to ethical consumption and rural development.

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## **Chapter 1: Introduction**

### **1.1 Background of Study**

The handmade and handicrafts sector represents one of the oldest and most culturally significant industries worldwide, particularly in India where artisanal traditions are deeply embedded in social and economic life. Handmade products are not only commodities but also cultural artefacts that reflect heritage, craftsmanship, and sustainability. Despite this rich legacy, the sector faces challenges in adapting to modern business environments characterized by globalization, digital transformation, and rapidly changing consumer preferences.

In recent years, consumer behaviour has shifted toward authenticity, eco-friendliness, and personalized experiences. Customers increasingly value products that carry cultural meaning and ethical production practices. This trend has created opportunities for handmade businesses to differentiate themselves from mass-produced goods. However, small enterprises often lack structured marketing strategies, advanced analytics, and technological tools to fully capitalize on these opportunities.

### **Rationale for the Study**

Small handmade businesses typically operate with limited resources, fragmented supply chains, and minimal exposure to advanced management practices. Traditional marketing approaches, largely based on demographic segmentation, fail to capture the complexity of consumer behaviour in this sector. For example, while age and gender provide basic insights, they do not explain deeper motivations such as lifestyle choices, cultural values, or sustainability concerns.

Artificial Intelligence (AI) offers a transformative solution by enabling businesses to analyse customer data more effectively, identify meaningful segments, and deliver personalized marketing strategies. AI-driven segmentation goes beyond demographics to incorporate behavioural, psychographic, and preference-based variables. This allows businesses to design campaigns that resonate more strongly with customers, thereby improving engagement, loyalty, and conversion rates.

### **Academic Context**

From an academic perspective, this study integrates management concepts such as customer segmentation, consumer behaviour analysis, and marketing strategy with modern technological applications. It demonstrates how theoretical frameworks can be applied to real-world business problems faced by small enterprises. The research contributes to existing literature by validating the role of AI in enhancing segmentation accuracy and marketing effectiveness, particularly in resource-constrained environments.

The study also aligns with broader themes in management education, including innovation, sustainability, and digital transformation. By focusing on handmade businesses, it highlights the intersection of tradition and technology, showing how modern tools can preserve cultural heritage while ensuring competitiveness in global markets.

### **Practical Context**

For handmade small businesses, the findings of this study provide a roadmap to adopt affordable AI tools without requiring extensive technical expertise. The research emphasizes practical strategies such as:

- Identifying customer segments based on education, lifestyle, and product preferences.
- Designing targeted campaigns for specific segments (e.g., price-sensitive youth, health-conscious professionals, premium buyers).
- Leveraging digital platforms to expand market reach and reduce dependence on intermediaries.
- Enhancing service quality and authenticity to strengthen customer loyalty.

By applying AI-driven segmentation, handmade businesses can overcome challenges such as limited budgets, fragmented markets, and competition from machine-made substitutes. The study demonstrates that even small enterprises can achieve scalability and sustainability by integrating modern marketing practices with traditional craftsmanship.

### **Importance of the Study**

- **Academic Importance:** The study contributes to management research by providing empirical evidence on the effectiveness of AI-driven segmentation in small business contexts. It bridges the gap between theoretical knowledge and practical application.

- **Practical Importance:** The findings offer actionable insights for artisans, entrepreneurs, and policymakers. They highlight affordable strategies that can empower small businesses to compete in digital marketplaces while preserving cultural heritage.
- **Societal Importance:** By supporting handmade businesses, the study promotes sustainable livelihoods, women's empowerment, and rural development. It also aligns with global trends toward ethical consumption and sustainable production.

## 1.2 Industry Profile

### Overview

The Indian handicrafts industry is one of the most significant sectors of the economy, deeply rooted in cultural heritage and artisanal traditions. It is a labour-intensive industry that provides employment to more than **seven million artisans**, with women comprising over **56% of the workforce**. The sector encompasses diverse product categories such as woodware, art metal wares, hand-printed textiles, embroidered goods, imitation jewellery, pottery, glassware, and home décor items.

### Market Size and Growth

- The Indian handicrafts market was valued at **US\$ 4.56 billion in 2024** and is projected to reach **US\$ 8.19 billion by 2033**, growing at a **CAGR of 6.39%**.
- India has **744 handicraft clusters**, employing nearly **212,000 artisans** and offering more than **35,000 unique products**.
- Major production clusters include **Surat, Bareilly, Varanasi, Agra, Hyderabad, Lucknow, Chennai, and Mumbai**, each specializing in distinct crafts.

### Export Potential

- Handicrafts exports reflect India's rich artisanal heritage and growing relevance in global home, lifestyle, and décor markets.
- Strong demand exists for categories such as **woodware, metal crafts, embroidered products, and textiles**.
- E-commerce platforms have opened new channels for artisans and MSMEs to access international markets.

## Government Support

- The **Development Commissioner (Handicrafts)** under the Ministry of Textiles provides schemes for skill development, marketing assistance, and financial support.
- National and international fairs, such as the **International Carpet Fair**, promote Indian handicrafts globally.

## Challenges

- **Fragmentation:** The industry is highly decentralized, with artisans spread across rural clusters.
- **Technology adoption:** Limited digital readiness among small artisans restricts access to modern marketing tools.
- **Competition:** Rising competition from machine-made substitutes and global imports.
- **Supply chain issues:** Dependence on middlemen often reduces artisans' profit margins.

## 1.3 Company Profile

### Introduction

The company selected for this study is Amazon, which serves as a major e-commerce platform supporting small-scale handmade enterprises through initiatives such as Amazon Handmade. Established in 1994, Amazon has grown into a global marketplace that enables artisans and small businesses to sell their handcrafted products to a wide customer base. Its operations are global, including a strong presence in India, where it connects local artisans with both domestic and international customers. By integrating traditional craftsmanship with modern marketing practices and AI-driven technologies, Amazon facilitates customer segmentation and personalized marketing for handmade product sellers.

### Nature of Business

- **Core Products:** Textiles, jewellery, pottery, organic food items, and home décor.
- **Target Market:** Middle- to upper-income customers who value authenticity, sustainability, and cultural heritage.

- **Business Model:** Direct-to-consumer sales through exhibitions, local stores, and e-commerce platforms.

### **Vision and Mission**

- **Vision:** To preserve traditional craftsmanship while adapting to modern consumer needs.
- **Mission:** To deliver authentic, sustainable, and high-quality handmade products that foster cultural appreciation and customer loyalty.

### **Organizational Structure**

- Small team of artisans supported by a management unit handling marketing, sales, and logistics.
- Family-owned and community-driven, with strong emphasis on ethical sourcing and fair wages for artisans.

### **Market Presence**

- Local presence through exhibitions and fairs.
- Growing online presence via social media and e-commerce platforms.
- Export potential explored through collaborations with international marketplaces.

### **Strengths**

- Strong cultural heritage and authenticity.
- Skilled artisans with unique product designs.
- Loyal customer base in niche markets.

### **Challenges**

- Limited financial and technological resources.
- Dependence on seasonal demand and exhibitions.
- Competition from machine-made substitutes and global imports.

## **1.4 Problem Statement**

### **Introduction**

The handmade and handicrafts sector, despite its cultural richness and economic potential, faces significant challenges in adapting to modern business environments. Small handmade enterprises often operate in fragmented markets, with limited access to structured customer insights, advanced marketing tools, and financial resources. While consumer demand for authentic, sustainable, and personalized products is rising, many handmade businesses struggle to position themselves competitively in the digital era.

This study identifies the core problems faced by handmade small businesses and explores how **AI-driven customer segmentation and personalized marketing** can provide practical solutions. The problem statement highlights the gap between traditional practices and modern requirements, forming the foundation for the research objectives.

## **Key Problems**

### **1. Limited Market Understanding**

- Handmade businesses often rely on intuition or anecdotal evidence to understand customer preferences.
- Traditional demographic segmentation (age, gender, income) fails to capture deeper behavioural and psychographic factors such as lifestyle, cultural values, and sustainability concerns.
- This lack of structured insights leads to ineffective marketing campaigns and missed opportunities.

### **2. Resource Constraints**

- Small enterprises typically operate with limited budgets, manpower, and technological infrastructure.
- They cannot afford large-scale market research or advanced analytics tools used by bigger corporations.
- As a result, marketing efforts remain generic and fail to resonate with diverse customer segments.

### **3. Competition from Mass-Produced Goods**

- Machine-made substitutes and imported products often dominate markets due to lower prices and wider availability.

- Handmade businesses struggle to differentiate themselves, despite offering authenticity and cultural value.
- Without targeted marketing, customers may overlook handmade products in favor of cheaper alternatives.

#### **4. Digital Adoption Challenges**

- Many artisans and small enterprises lack digital literacy and readiness to adopt modern tools.
- While e-commerce and social media platforms provide opportunities, businesses often fail to leverage them effectively.
- This results in limited online visibility and restricted access to broader markets.

#### **5. Customer Engagement Gaps**

- Repeat purchases and loyalty are strongly influenced by service quality and personalization.
- Handmade businesses often lack mechanisms to deliver tailored experiences, loyalty programs, or personalized offers.
- This reduces customer retention and long-term profitability.

### **Research Gap**

Although AI-driven segmentation and personalized marketing have been widely adopted in larger organizations, their application in small handmade businesses remains underexplored. Existing studies focus on general marketing strategies or traditional segmentation methods, leaving a gap in understanding how affordable AI tools can empower resource-constrained enterprises. This research seeks to fill that gap by demonstrating practical, actionable strategies tailored to the handmade sector.

### **Problem Statement**

Handmade small businesses face a critical challenge in balancing tradition with modern competitiveness. They lack structured customer insights, operate with limited resources, and struggle to differentiate themselves in markets dominated by mass-produced

goods. Their inability to adopt advanced marketing tools results in generic campaigns, weak customer engagement, and restricted growth.

The central problem addressed in this study is: **How can AI-driven customer segmentation and personalized marketing strategies be effectively applied in handmade small businesses to overcome resource constraints, improve customer engagement, and achieve sustainable growth?**

## **1.5 Need of Study**

### **Introduction**

The handmade and handicrafts sector is a vital part of the economy, contributing to employment generation, cultural preservation, and sustainable development. Despite its importance, small handmade businesses face persistent challenges in marketing, customer engagement, and competitiveness. Traditional marketing methods often fail to capture the complexity of consumer behaviour, while limited resources restrict the adoption of advanced tools.

In this context, the need for this study arises from the growing importance of **artificial intelligence (AI)-driven customer segmentation and personalized marketing** as practical solutions for small enterprises. The study aims to demonstrate how management concepts can be applied to real-world problems, providing actionable strategies that empower handmade businesses to thrive in competitive markets.

### **Academic Need**

#### **1. Bridging Theory and Practice**

- MBA education emphasizes the application of management concepts to real business problems.
- This study integrates theories of customer segmentation, consumer behavior, and marketing strategy with modern AI applications.
- It contributes to academic literature by validating the relevance of AI in resource-constrained environments.

#### **2. Expanding Research Scope**

- Existing studies focus primarily on large corporations or technology-driven industries.
- There is limited research on how AI can be applied in small handmade businesses.
- This study fills that gap by providing empirical evidence and case-based insights.

### **3. Contribution to Management Knowledge**

- The study enhances understanding of how segmentation accuracy influences marketing effectiveness.
- It demonstrates the role of AI in improving decision-making, customer engagement, and business sustainability.

## **Practical Need**

### **1. Empowering Small Businesses**

- Handmade enterprises often lack structured customer insights.
- AI-driven segmentation provides affordable tools to identify meaningful customer groups and design targeted campaigns.
- This enables businesses to compete effectively against mass-produced goods.

### **2. Improving Customer Engagement**

- Personalized marketing enhances customer satisfaction, loyalty, and repeat purchases.
- By tailoring offers and communication, businesses can strengthen relationships with customers.

### **3. Leveraging Digital Platforms**

- E-commerce and social media have opened new opportunities for handmade businesses.
- AI tools can help optimize digital marketing strategies, improve visibility, and expand market reach.

#### **4. Sustainability and Cultural Preservation**

- Supporting handmade businesses contributes to sustainable livelihoods, women's empowerment, and rural development.
- By adopting modern tools, artisans can preserve cultural heritage while ensuring economic viability.

### **Managerial Need**

#### **1. Decision-Making Support**

- Managers in small enterprises often rely on intuition rather than data-driven insights.
  - AI-driven segmentation provides structured information to support strategic decisions.

#### **2. Resource Optimization**

- Limited budgets require efficient allocation of marketing resources.
- Targeted campaigns reduce waste and maximize impact.

#### **3. Competitive Advantage**

- Differentiation through authenticity and personalization helps businesses stand out in crowded markets.
- AI tools provide a sustainable competitive edge by aligning products with customer expectations.

### **Societal Need**

#### **1. Employment Generation**

- The handmade sector employs millions of artisans, many from marginalized communities.
- Strengthening small businesses ensures continued livelihood opportunities.

#### **2. Ethical Consumption**

- Consumers increasingly demand sustainable and ethically produced goods.

- Handmade businesses align with these values, and effective marketing can amplify their appeal.

### 3. Rural Development

- Many handmade enterprises are located in rural areas.
- Supporting them contributes to balanced regional development and reduces urban migration.

## 1.6 Objectives of the Study

### Introduction

Every academic research project must be guided by clear objectives that define its scope, direction, and expected outcomes. Objectives serve as the foundation for the study, ensuring that the research remains focused, systematic, and aligned with both academic and practical requirements. In the context of this project, which explores **AI-driven customer segmentation and personalized marketing for handmade small businesses**, the objectives are designed to bridge the gap between management theory and real-world application.

The handmade sector faces unique challenges such as resource constraints, fragmented markets, and limited technological adoption. Therefore, the objectives of this study are structured to address these challenges while demonstrating the relevance of modern management concepts and tools.

### Primary Objective

The **primary objective** of this study is:

- **To analyse the role of AI-driven customer segmentation and personalized marketing in enhancing competitiveness, customer engagement, and sustainable growth of handmade small businesses.**

This overarching objective provides the central focus of the research and guides the formulation of specific secondary objectives.

### Secondary Objectives

1. **To examine the current marketing practices of handmade small businesses.**
  - Identify strengths and weaknesses in traditional approaches.

- Assess the reliance on demographic segmentation and its limitations.
- 2. To evaluate customer preferences and purchase behaviour in the handmade sector.**
- Understand demographic, psychographic, and behavioural factors influencing buying decisions.
  - Highlight the importance of authenticity, sustainability, and personalization.
- 3. To explore the applicability of AI tools in customer segmentation.**
- Demonstrate how AI can analyse complex datasets to identify meaningful customer groups.
  - Compare AI-driven segmentation with traditional methods in terms of accuracy and relevance.
- 4. To design personalized marketing strategies based on AI-driven insights.**
- Develop targeted campaigns for different customer segments.
  - Suggest affordable and practical tools that small businesses can adopt.
- 5. To assess the impact of personalized marketing on customer engagement and loyalty.**
- Measure how tailored offers and communication influence repeat purchases.
  - Evaluate customer satisfaction and long-term retention.
- 6. To provide actionable recommendations for handmade businesses.**
- Suggest strategies for integrating AI tools into daily operations.
  - Highlight best practices for resource optimization and competitive differentiation.
- 7. To contribute to academic literature on AI in small business marketing.**
- Validate management concepts in the context of resource-constrained enterprises.
  - Expand research scope beyond large corporations to include traditional industries.

## Importance of Objectives

The objectives of this study are significant for several reasons:

- **Academic Relevance:** They ensure that the research contributes to management knowledge by applying theories of segmentation, consumer behaviour, and marketing strategy in a new context.
- **Practical Utility:** They provide small businesses with a roadmap to adopt affordable AI tools and improve competitiveness.
- **Managerial Guidance:** They support decision-making by offering structured insights into customer behaviour and marketing effectiveness.
- **Societal Impact:** They promote sustainable livelihoods, cultural preservation, and ethical consumption by strengthening handmade enterprises.

## Alignment with Research Framework

The objectives are closely aligned with the overall research framework:

- **Chapter 1 (Introduction)** establishes the background, industry profile, company profile, and problem statement.
- **Chapter 2 (Literature Review)** connects the objectives to existing theories and prior studies.
- **Chapter 3 (Methodology)** outlines the research design to achieve these objectives.
- **Chapter 4–6 (Findings, Analysis, and Conclusions)** present results that directly address the objectives.
- **Chapter 7–8 (Recommendations and Limitations)** highlights practical implications and areas for improvement.

## 1.7 Research Questions / Hypothesis

### Introduction

Research questions and hypotheses form the backbone of any academic study. They provide clarity, direction, and focus, ensuring that the investigation remains systematic and purposeful. In the context of this project, which explores **AI-driven customer segmentation and personalized marketing for handmade small businesses**, the research questions are

designed to address the challenges identified in the problem statement, while the hypotheses provide testable propositions that can be validated through empirical analysis.

This section outlines the key research questions and hypotheses that guide the study, ensuring alignment with the objectives and scope of the research.

## **Research Questions**

The following research questions have been formulated to guide the study:

### **1. Customer Segmentation**

- How do demographic, psychographic, and behavioural factors influence customer preferences for handmade products?
- Can AI-driven segmentation provide more accurate and meaningful customer groups compared to traditional methods?

### **2. Marketing Effectiveness**

- What role does personalized marketing play in enhancing customer engagement and loyalty in handmade businesses?
- How do customers respond to tailored offers and communication compared to generic marketing campaigns?

### **3. Technology Adoption**

- What are the barriers and enablers for adopting AI tools in small handmade enterprises?
- How can affordable AI solutions be integrated into daily business operations without requiring extensive technical expertise?

### **4. Business Impact**

- To what extent can AI-driven segmentation and personalized marketing improve sales, customer retention, and overall competitiveness of handmade businesses?
- What strategies can be recommended to ensure sustainable growth while preserving cultural heritage?

## Hypotheses

Based on the research questions, the following hypotheses have been developed:

**H1: AI-driven customer segmentation provides more accurate and actionable insights than traditional demographic segmentation methods.**

- Rationale: Traditional segmentation often overlooks behavioural and psychographic variables, whereas AI can analyse complex datasets to identify meaningful patterns.

**H2: Personalized marketing strategies based on AI-driven insights significantly enhance customer engagement and loyalty compared to generic campaigns.**

- Rationale: Tailored communication resonates more strongly with customers, leading to higher satisfaction and repeat purchases.

**H3: Adoption of AI tools in handmade small businesses is positively correlated with improved competitiveness and market reach.**

- Rationale: Businesses that integrate AI into their operations can optimize resources, differentiate themselves, and expand into digital marketplaces.

**H4: Customers of handmade products demonstrate higher purchase intention when marketing emphasizes authenticity, sustainability, and personalization.**

- Rationale: Consumer trends indicate growing demand for ethical and culturally meaningful products, which can be amplified through targeted marketing.

**H5: Resource constraints (financial, technological, and human) act as significant barriers to AI adoption in handmade small businesses.**

- Rationale: Limited budgets and digital literacy often restrict small enterprises from leveraging advanced tools, requiring affordable and simplified solutions.

## Importance of Research Questions and Hypotheses

- **Academic Importance:** They ensure that the study contributes to management literature by validating theories in a new context.
- **Practical Importance:** They provide small businesses with evidence-based insights into customer behaviour and marketing effectiveness.

- **Managerial Importance:** They guide decision-making by offering structured propositions that can be tested and applied in real-world scenarios.
- **Societal Importance:** They highlight the role of handmade businesses in promoting sustainability, cultural preservation, and rural development.

## 1.8 Scope of the Study

### Introduction

The scope of a study defines its boundaries, coverage, and limitations. It clarifies what the research will include, what it will exclude, and how far its findings can be generalized. In the context of this project, which explores **AI-driven customer segmentation and personalized marketing for handmade small businesses**, the scope is designed to ensure academic rigor while maintaining practical relevance.

This section outlines the **geographic scope, industry scope, methodological scope, and conceptual scope** of the study, along with its limitations and areas of applicability.

### Geographic Scope

- The study focuses primarily on handmade businesses operating in **India**, with particular emphasis on small enterprises located in semi-urban and rural clusters.
- Data collection is limited to selected regions where handmade products are widely produced and consumed.
- While the findings may have relevance for other countries with similar artisanal industries, the primary context remains India.

### Industry Scope

- The research is confined to the **handmade and handicrafts sector**, including product categories such as textiles, jewellery, pottery, organic food items, and home décor.
- Large-scale manufacturing units and machine-made substitutes are excluded from the study.
- The emphasis is on **small and medium-sized enterprises (SMEs)** that rely on traditional craftsmanship and face resource constraints.

### Methodological Scope

- The study employs a **mixed-method approach**, combining quantitative surveys with qualitative insights.
- Primary data is collected through questionnaires administered to customers of handmade products.
- Secondary data is drawn from academic literature, industry reports, and government publications.
- Statistical tools such as correlation, regression, and cluster analysis are used to validate hypotheses and derive insights.
- The study is limited to a sample size of approximately **150 respondents**, which provides meaningful insights but restricts generalizability.

### Conceptual Scope

- The research focuses on **customer segmentation and personalized marketing** as key management concepts.
- Segmentation variables include demographic (age, gender, education), behavioural (purchase frequency, payment methods), and psychographic (lifestyle, sustainability preferences).
- Personalized marketing strategies are examined in terms of tailored offers, communication channels, and customer engagement mechanisms.
- The role of **AI tools** is explored as a means of enhancing segmentation accuracy and marketing effectiveness.
- Other aspects of business management, such as supply chain optimization or financial management, are outside the scope of this study.

### Academic Scope

- The study contributes to management literature by applying theoretical frameworks to a practical, resource-constrained context.
- It validates the relevance of AI in small business marketing, an area that has received limited academic attention.

- The findings are intended to serve as a reference for future research on digital transformation in traditional industries.

### **Practical Scope**

- The research provides actionable recommendations for handmade businesses seeking to adopt affordable AI tools.
- It highlights strategies for improving customer engagement, loyalty, and competitiveness.
- The study is designed to be useful for entrepreneurs, managers, policymakers, and industry stakeholders.
- While the recommendations are tailored to handmade businesses, they may also be applicable to other small enterprises facing similar challenges.

### **Limitations of Scope**

- The study is limited by its sample size and geographic coverage, which restricts the generalizability of findings.
- Resource constraints prevent the inclusion of longitudinal data or large-scale experiments.
- The focus on AI-driven segmentation excludes other potential marketing innovations such as influencer marketing or omnichannel strategies.
- Company-specific variations in resources, digital readiness, and customer base may affect the applicability of recommendations.

## Chapter 2: Literature Review

### 2.1 Theoretical Concepts

#### Introduction

The theoretical foundation of any research project provides the intellectual framework within which the study is conducted. It ensures that the research is not only practical but also academically rigorous, grounded in established theories and models. In the context of this project, which explores **AI-driven customer segmentation and personalized marketing for handmade small businesses**, several management and marketing theories are relevant. These concepts guide the research design, data analysis, and interpretation of findings.

#### Customer Segmentation Theory

Customer segmentation is a fundamental concept in marketing that involves dividing a heterogeneous market into smaller, homogeneous groups of consumers with similar needs, preferences, or behaviours.

- **Demographic Segmentation:** Based on variables such as age, gender, income, and education.
- **Psychographic Segmentation:** Focuses on lifestyle, values, and personality traits.
- **Behavioural Segmentation:** Examines purchase frequency, brand loyalty, and usage patterns.
- **Geographic Segmentation:** Considers location, culture, and regional preferences.

The theoretical basis for segmentation lies in the principle that **different customer groups respond differently to marketing strategies**. By identifying these groups, businesses can tailor their offerings and communication to maximize effectiveness.

#### Consumer Behaviour Theory

Consumer behaviour theory explains how individuals make decisions about purchasing products and services. It integrates psychological, social, and cultural factors.

- **Maslow's Hierarchy of Needs:** Suggests that consumer choices are influenced by the desire to satisfy needs ranging from basic (physiological) to advanced (self-actualization). Handmade products often appeal to esteem and self-actualization needs, as they symbolize uniqueness and cultural identity.

- **Theory of Planned Behaviour (Ajzen, 1991):** Proposes that attitudes, subjective norms, and perceived behavioural control influence purchase intentions. This is relevant in understanding why consumers choose handmade products over mass-produced alternatives.
- **SERVQUAL Model (Parasuraman et al., 1988):** Highlights the importance of service quality dimensions—reliability, responsiveness, assurance, empathy, and tangibles—in shaping customer satisfaction and loyalty.

### Marketing Strategy Theories

Marketing strategy theories provide frameworks for positioning products and designing campaigns.

- **Porter's Generic Strategies:** Handmade businesses often rely on **differentiation**, emphasizing authenticity, craftsmanship, and sustainability to stand out from mass-produced goods.
- **Kotler's Marketing Mix (4Ps):** Product, Price, Place, and Promotion remain central to marketing handmade products. AI-driven segmentation enhances the effectiveness of each element by aligning them with customer preferences.
- **Relationship Marketing Theory:** Focuses on building long-term relationships rather than short-term transactions. Personalized marketing strengthens customer loyalty by fostering trust and emotional connection.

### Artificial Intelligence in Marketing

AI introduces a new dimension to traditional marketing theories by enabling advanced data analysis and personalization.

- **Machine Learning Models:** Algorithms such as clustering (K-Means) and classification help identify customer segments based on complex datasets.
- **Predictive Analytics:** AI can forecast customer behaviour, enabling proactive marketing strategies.
- **Personalization Theory:** Suggests that tailored communication enhances customer engagement and satisfaction. AI operationalizes this theory by delivering customized offers and recommendations.

## Decision-Making Theories

Decision-making theories explain how managers and businesses make strategic choices.

- **Rational Decision-Making Model:** Assumes that decisions are made logically based on data and analysis. AI-driven insights support this model by providing structured information.
- **Bounded Rationality (Simon, 1957):** Recognizes that decisions are limited by constraints such as time, resources, and information. AI helps overcome these limitations by processing large datasets efficiently.
- **Evidence-Based Management:** Emphasizes the use of empirical data and scientific methods in decision-making. This study aligns with this theory by applying AI-driven analysis to real business problems.

## Integration of Theories

The theoretical concepts discussed above are interconnected:

- **Segmentation theory** provides the foundation for identifying customer groups.
- **Consumer behaviour theory** explains the motivations behind purchasing decisions.
- **Marketing strategy theories** guide the design of campaigns tailored to specific segments.
- **AI applications** operationalize these theories by enhancing accuracy and personalization.
- **Decision-making theories** ensure that insights are translated into actionable strategies.

Together, these concepts create a comprehensive framework for analysing the role of AI-driven segmentation and personalized marketing in handmade small businesses.

## 2.2 Models & Frameworks

### Introduction

Models and frameworks provide structured approaches to analyzing business problems and designing solutions. They serve as guiding tools that ensure research is systematic, evidence-based, and aligned with established management practices. In the

context of this study, which explores **AI-driven customer segmentation and personalized marketing for handmade small businesses**, several models and frameworks are particularly relevant. These include classical marketing models, customer segmentation frameworks, and modern AI-based analytical models.

## **Marketing Models**

### **1. Kotler's Marketing Mix (4Ps)**

- **Product:** Handmade products emphasize authenticity, craftsmanship, and cultural heritage.
- **Price:** Pricing strategies must balance affordability with the premium value of artisanal goods.
- **Place:** Distribution through exhibitions, local stores, and digital platforms.
- **Promotion:** Marketing campaigns tailored to customer segments using AI-driven insights.
- **Relevance:** AI enhances the effectiveness of the 4Ps by aligning them with customer preferences.

### **2. Porter's Generic Strategies**

- **Cost Leadership:** Not applicable for handmade businesses due to higher production costs.
- **Differentiation:** Handmade enterprises rely on uniqueness, authenticity, and sustainability.
- **Focus Strategy:** Targeting niche markets such as eco-conscious consumers or cultural enthusiasts.
- **Relevance:** AI-driven segmentation strengthens differentiation and focus strategies by identifying niche customer groups.

### **3. Relationship Marketing Framework**

- Emphasizes building long-term customer relationships rather than short-term transactions.
- Personalized marketing fosters trust, loyalty, and repeat purchases.

- Relevance: AI enables businesses to design loyalty programs and tailored communication strategies.

## **Customer Segmentation Frameworks**

### **1. STP Model (Segmentation, Targeting, Positioning)**

- **Segmentation:** Dividing the market into distinct groups based on demographics, psychographics, and behaviour.
- **Targeting:** Selecting the most attractive segments to serve.
- **Positioning:** Designing marketing strategies to occupy a distinct place in the minds of customers.
- Relevance: AI enhances segmentation accuracy, enabling more effective targeting and positioning.

### **2. RFM Model (Recency, Frequency, Monetary Value)**

- **Recency:** How recently a customer made a purchase.
- **Frequency:** How often a customer buys.
- **Monetary Value:** How much a customer spends.
- Relevance: AI can automate RFM analysis, identifying high-value customers for personalized campaigns.

### **3. Customer Lifetime Value (CLV) Framework**

- Estimates the total value a customer contributes over their relationship with the business.
- Helps prioritize marketing efforts toward high-value customers.
- Relevance: AI-driven predictive analytics improve CLV estimation and resource allocation.

## **AI-Based Analytical Models**

### **1. Clustering Models (K-Means, Hierarchical Clustering)**

- Group customers into clusters based on similarities in preferences and behaviour.

- Relevance: Identifies hidden patterns in customer data, enabling precise segmentation.

## 2. Classification Models (Decision Trees, Random Forests)

- Predict customer behaviour based on historical data.
- Relevance: Helps forecast purchase intentions and design proactive marketing strategies.

## 3. Predictive Analytics Framework

- Uses machine learning algorithms to forecast future trends.
- Relevance: Enables businesses to anticipate customer needs and design personalized offers.

## 4. Personalization Models

- AI algorithms recommend products and offer tailored to individual customers.
- Relevance: Enhances customer satisfaction and loyalty by delivering unique experiences.

## Integrated Framework for Handmade Businesses

The study integrates traditional marketing models with modern AI-based frameworks to create a holistic approach:

- **STP + AI Clustering:** Enhances segmentation accuracy.
- **4Ps + Personalization Models:** Aligns product, price, place, and promotion with customer preferences.
- **RFM + Predictive Analytics:** Identifies high-value customers and forecasts future behaviour.
- **Relationship Marketing + AI Personalization:** Builds long-term loyalty through tailored communication.

This integrated framework ensures that handmade businesses can leverage both classical theories and modern technologies to achieve sustainable growth.

## 2.3 Review of Previous Studies

Each study is summarized with **author, year, objective, methodology, findings, and limitations**:

- **Huang & Wang (2020)**: Reviewed AI applications in customer segmentation.
  - *Objective*: To assess how AI enhances personalization.
  - *Methodology*: Literature synthesis of AI models in marketing.
  - *Findings*: AI improves accuracy of segmentation and personalization.
  - *Limitation*: Focused on large enterprises; limited insights for SMEs.
- **Preethi et al. (2025)**: Studied AI-powered approaches in small-scale businesses.
  - *Objective*: To evaluate AI's role in sustainable marketing for micro-enterprises.
  - *Methodology*: Case studies of small businesses in India.
  - *Findings*: AI tools improved targeting and customer retention.
  - *Limitation*: Small sample size; limited generalizability.
- **Le Dinh et al. (2025)**: Examined AI in SMEs.
  - *Objective*: To explore AI's role in enhancing business functions.
  - *Methodology*: Survey of SMEs across Canada and Vietnam.
  - *Findings*: AI adoption improved efficiency but faced barriers like cost and expertise.
  - *Limitation*: Did not focus specifically on marketing.
- **Castelli et al. (2025)**: Investigated AI personalization in SMEs.
  - *Objective*: To analyse how SMEs integrate AI into segmentation.
  - *Methodology*: Case-based analysis of European SMEs.
  - *Findings*: AI improved personalization but required cultural and organizational adaptation.
  - *Limitation*: Limited focus on handmade businesses.

- **Moqaddemi (2025):** Reviewed AI's impact on personalization and engagement.
  - *Objective:* To assess hyper-personalized marketing strategies.
  - *Methodology:* Critical literature review of 2020–2025 studies.
  - *Findings:* AI enables real-time engagement and predictive analytics.
  - *Limitation:* Ethical concerns around manipulation and privacy .

## 2.4 Research Gap Identification

### Introduction

Identifying research gaps is a critical step in academic inquiry. It highlights areas where existing literature is insufficient, outdated, or lacks practical application. Research gaps provide justification for the study and demonstrate its originality and contribution to knowledge. In the context of this project, which explores **AI-driven customer segmentation and personalized marketing for handmade small businesses**, the identification of gaps ensures that the research addresses unmet needs in both academic and practical domains.

### Gaps in Customer Segmentation Research

#### 1. Overemphasis on Demographics

- Most existing studies on customer segmentation in small businesses rely heavily on demographic variables such as age, gender, and income.
- These approaches fail to capture deeper behavioural and psychographic factors, such as lifestyle, cultural values, and sustainability preferences, which are highly relevant in the handmade sector.

#### 2. Limited Application in Traditional Industries

- While segmentation has been extensively studied in large corporations and technology-driven industries, its application in traditional, resource-constrained sectors like handicrafts remains underexplored.
- There is a lack of empirical evidence on how segmentation can be tailored to artisanal businesses.

#### 3. Neglect of AI Tools

- Existing literature often discusses segmentation in theoretical terms but rarely examines the role of AI in enhancing segmentation accuracy.
- Few studies demonstrate how affordable AI tools can be practically adopted by small enterprises.

## **Gaps in Personalized Marketing Research**

### **1. Generic Marketing Approaches**

- Many handmade businesses continue to rely on generic marketing campaigns that fail to resonate with diverse customer segments.
- Academic studies have not sufficiently addressed how personalized marketing can be adapted to small enterprises with limited resources.

### **2. Lack of Empirical Evidence**

- While personalization is widely recognized as effective in e-commerce and large-scale retail, there is limited empirical research on its impact in the handmade sector.
- The absence of case studies or field experiments leaves a gap in understanding its practical relevance.

### **3. Technology Adoption Challenges**

- Research often assumes that businesses have the capacity to adopt advanced tools.
- There is little exploration of barriers such as digital literacy, financial constraints, and infrastructural limitations faced by small handmade enterprises.

## **Gaps in AI and Small Business Research**

### **1. Focus on Large Enterprises**

- Most AI-related studies focus on multinational corporations or technology-intensive industries.
- Small businesses, particularly in traditional sectors, are often overlooked despite their potential to benefit from AI.

## **2. Affordability and Accessibility**

- Existing research rarely addresses how AI tools can be made affordable and accessible to resource-constrained enterprises.
- There is a need to explore simplified, cost-effective solutions tailored to small businesses.

## **3. Integration with Management Concepts**

- Few studies integrate AI applications with established management theories such as segmentation, consumer behaviour, and marketing strategy.
- This creates a gap in bridging academic frameworks with practical technological solutions.

## **Practical Gaps in Handmade Business Research**

### **1. Limited Market Insights**

- Handmade businesses often lack structured data on customer preferences and purchase behaviour.
- Research has not adequately addressed how small enterprises can collect and analyse customer data effectively.

### **2. Weak Digital Presence**

- While e-commerce platforms provide opportunities, many handmade businesses struggle to establish a strong digital presence.
- There is limited research on strategies to enhance online visibility and engagement for artisanal enterprises.

### **3. Sustainability and Cultural Preservation**

- Existing studies focus on economic aspects but often neglect the role of handmade businesses in promoting sustainability, women's empowerment, and cultural heritage.
- There is a need to explore how modern marketing tools can support these broader societal goals.

## Summary of Research Gaps

The review of existing literature reveals several critical gaps:

- Overreliance on demographic segmentation without considering behavioural and psychographic factors.
- Limited application of segmentation and personalization in traditional, resource-constrained industries.
- Lack of empirical evidence on the impact of personalized marketing in handmade businesses.
- Insufficient focus on affordability, accessibility, and integration of AI tools for small enterprises.
- Neglect of broader societal dimensions such as sustainability and cultural preservation.

By addressing these gaps, the study contributes to both academic knowledge and practical solutions. It demonstrates how AI-driven customer segmentation and personalized marketing can empower handmade small businesses to achieve competitiveness, sustainability, and customer-centric growth.

## Chapter 3: Research Methodology

### 3.1 Research Design

The research design provides the overall framework for conducting the study and ensures that the objectives are addressed systematically. For this project, a **descriptive research design** has been adopted.

A descriptive design is appropriate because the study seeks to **analyse existing customer characteristics, preferences, and behaviours** within handmade small businesses, rather than exploring entirely new phenomena or establishing strict cause-effect relationships. The purpose is to describe patterns and associations that can inform the development of AI-driven segmentation models and personalized marketing strategies.

#### Why Descriptive Design?

- **Exploratory research** is typically used when little is known about a subject and the aim is to generate hypotheses. While handmade businesses are under-researched in terms of AI adoption, there is already a body of literature on customer segmentation and personalization. Thus, exploratory design alone would be insufficient.
- **Causal research** focuses on establishing cause-and-effect relationships. While AI segmentation may influence marketing outcomes, this study does not attempt to prove direct causality but rather to identify associations and patterns.
- **Descriptive research** allows for systematic collection and analysis of data to describe customer demographics, preferences, and perceptions. It provides a structured basis for applying statistical techniques and machine learning models.

#### Application in This Study

- The design enables the collection of **primary data** through surveys and interviews, capturing customer demographics (e.g., age, gender, education) and preferences (e.g., product categories, purchase frequency).
- It supports the use of **secondary data** from academic literature and industry reports to contextualize findings.
- Statistical techniques such as correlation, regression, and chi-square tests are applied to identify relationships between variables.

- Machine learning clustering models (e.g., K-Means) are used to segment customers into distinct groups, providing actionable insights for personalized marketing.

### **Advantages of Descriptive Design in This Context**

- Provides **quantitative insights** into customer behaviour.
- Facilitates **comparisons across segments**, such as educational background and product preferences.
- Ensures **replicability and reliability**, as structured questionnaires and standardized tools are used.
- Bridges the gap between **academic rigor and practical application**, making findings useful for handmade small businesses.

In summary, the descriptive research design ensures that the study remains systematic, evidence-based, and aligned with the objectives of analysing customer segmentation and personalization strategies through AI. It provides the necessary foundation for subsequent chapters on data collection, analysis, and interpretation.

### **3.2 Type of Data**

The credibility of any research project depends on the quality and relevance of the data collected. For this study on *AI-driven customer segmentation and personalized marketing for handmade small businesses*, both **primary data** and **secondary data** were utilized to ensure a comprehensive understanding of the research problem.

#### **Primary Data**

Primary data refers to information collected directly from the source for the specific purpose of this study. In this project, primary data will be gathered from **customers of handmade small businesses in Chennai**.

- **Surveys:** Structured questionnaires were distributed to customers of handmade small businesses through online forms and physical copies at local exhibitions. These surveys captured quantitative data such as age, gender, educational background, product preferences, and purchase frequency.

- **Interviews:** Semi-structured interviews were conducted with business owners and artisans to understand their challenges in marketing, their awareness of AI tools, and their expectations from customer segmentation.
- **Observations:** Customer interactions at craft fairs and online platforms were observed to identify behavioural patterns, such as browsing habits, product inquiries, and decision-making processes.

**Rationale:** Primary data ensures relevance and accuracy, as it directly reflects the behaviour and preferences of the target population. It also allows the study to test hypotheses about education-based segmentation and its impact on customer loyalty.

### Secondary Data

Secondary data refers to information collected from existing sources to provide theoretical and contextual support for the study. It complements primary data by offering broader perspectives and established knowledge.

- **Industry Reports:** Publications on the Indian handicraft sector, highlighting market size, growth trends, and consumer behaviour.
- **Academic Journals:** Peer-reviewed studies on AI-driven segmentation, personalization, and SME adoption of technology.
- **Government Publications:** Data from ministries and trade bodies on employment, exports, and policies supporting handmade businesses.
- **Company Records:** Internal sales data, customer feedback, and marketing campaign reports from participating businesses.
- **Case Studies:** Documented examples of small businesses that have adopted AI tools were analysed to understand practical applications and limitations.
- **Academic Literature:** Research papers, journal articles, and books on AI in marketing, customer segmentation, and small business strategies were reviewed to build the theoretical framework.

**Rationale:** Secondary data strengthens the study by providing a broader context, enabling comparison with existing literature, and validating primary findings. It also helps identify industry-wide trends that may influence handmade businesses.

## Integration of Primary and Secondary Data

The combination of primary and secondary data ensures a **comprehensive methodology**:

- Primary data provides **first-hand insights** into customer behaviour and preferences.
- Secondary data offers **contextual depth**, linking findings to broader industry and academic perspectives.
- Together, they enhance the **validity and reliability** of the study, ensuring that conclusions are both evidence-based and practically relevant.
- Allowing **triangulation**, where findings from different sources are compared for consistency.
- Providing both **quantitative evidence** (surveys, statistical analysis) and **qualitative insights** (interviews, literature review).

### 3.3 Data Collection Methods

The success of any research project depends on the appropriateness and rigor of its data collection methods. For this study on *AI-driven customer segmentation and personalized marketing for handmade small businesses*, a combination of **survey, interview, and observation** methods was employed. This triangulated approach ensures both quantitative and qualitative insights, enhancing the reliability and validity of the findings.

#### Survey

Surveys were the primary method of data collection, designed to capture structured, quantitative information from customers of handmade small businesses. A questionnaire was developed with closed-ended and Likert-scale questions to measure:

Surveys are the primary tool for collecting quantitative data in this study.

- **Design:** Structured questionnaires with both closed-ended and Likert-scale questions.
- **Focus Areas:**
  - Demographic details (age, gender, education level).
  - Purchasing behaviour (frequency, spending patterns).
  - Preferences (authenticity, sustainability, price sensitivity).

- Perceptions of personalized marketing campaigns.
- **Distribution:** Online (Google Forms, email) and offline (printed forms in local handmade stores).
- **Advantages:**
  - Cost-effective and scalable.
  - Provides measurable data for statistical analysis.
- **Limitations:**
  - Risk of response bias.
  - Requires careful design to ensure clarity and avoid misinterpretation.

**Example Question:** *“On a scale of 1–5, how important is product authenticity when purchasing handmade goods?”*

### Interview

Semi-structured interviews were conducted with business owners and artisans to gain qualitative insights into their marketing practices and challenges. The interviews focused on:

- **Participants:** Selected customers and business owners of handmade small businesses in Chennai.
- **Format:** Semi-structured interviews, allowing flexibility while maintaining focus on key themes.
- **Focus Areas:**
  - Customer perceptions of personalized marketing.
  - Business owners’ challenges in adopting AI-driven segmentation.
  - Expectations regarding education-based targeting.
- **Advantages:**
  - Provides depth and context to survey findings.
  - Captures emotions, attitudes, and motivations that quantitative data may miss.
- **Limitations:**

- Time-consuming.
- Smaller sample size compared to surveys, limiting generalizability.

**Example Interview Prompt:** *“Can you describe how your education level influences your choice of handmade products or your response to marketing campaigns?”*

## Observation

Observation is employed to understand customer behaviour in natural settings.

- **Scope:**
  - Physical stores: Observing customer interactions with products, staff, and promotional materials.
  - Online platforms: Tracking browsing behaviour, product clicks, and engagement with personalized recommendations.
- **Advantages:**
  - Provides unbiased, real-time insights into customer behaviour.
  - Helps validate survey and interview responses.
- **Limitations:**
  - Observer bias may occur.
  - Ethical considerations in online tracking (requires consent).

**Example:** Observing whether customers with higher education levels spend more time reading product descriptions or sustainability labels compared to others.

## Integration of Methods

By combining surveys, interviews, and observations, the study achieved **methodological triangulation**, which enhances the robustness of findings. Surveys provided quantitative data for statistical analysis, interviews offered qualitative depth, and observations validated behavioural patterns. Together, these methods ensured a holistic understanding of customer segmentation and personalized marketing in handmade small businesses.

- Surveys provide **quantitative breadth**.
- Interviews offer **qualitative depth**.

- Observations validate findings through **real-world behaviour**.

This triangulation enhances the **reliability and validity** of the research outcomes.

### 3.4 Sampling Design

Sampling design is a critical component of research methodology, as it determines the representativeness and reliability of the data collected. For this study on *AI-driven customer segmentation and personalized marketing for handmade small businesses*, the sampling design was carefully structured to ensure that the findings reflect the diversity of customers and provide actionable insights for small enterprises.

#### Population

The population for this study consists of **customers of handmade small businesses**, both online and offline. This includes individuals who purchase artisanal products such as textiles, jewellery, pottery, organic food, and home décor. The population is diverse in terms of age, gender, educational background, and purchasing behaviour.

Since handmade businesses often operate in niche markets, the population is not limited to a specific geographic region but includes customers from local craft fairs, exhibitions, and digital platforms such as Etsy, Amazon Handmade, and Instagram marketplaces. This broad definition ensures that the study captures the perspectives of both traditional buyers and digitally active consumers.

The population for this study consists of **customers of handmade small businesses in Chennai, India**.

- These businesses include artisans, craft shops, and small enterprises producing handmade goods such as textiles, jewellery, pottery, and decorative items.
- The population is diverse, spanning different age groups, income levels, and educational backgrounds.
- Since the study focuses on **education-based segmentation**, the population is categorized into strata based on education level (e.g., high school, undergraduate, postgraduate).

**Rationale:** This ensures that the study captures how education influences purchasing behaviour and responsiveness to personalized marketing.

## Sample Size

A total of **150 respondents** were selected for the study. This sample size was chosen based on the following considerations

- **Feasibility:** Handmade businesses typically have limited customer bases, making very large samples impractical.
- **Statistical reliability:** A sample of 150 provides sufficient data for applying statistical techniques such as correlation, regression, and chi-square tests.
- **Balance between depth and breadth:** The sample size allows for meaningful segmentation while remaining manageable for analysis.
- **Justification:**
  - A sample of this size is sufficient to apply statistical techniques such as correlation, regression, and chi-square tests.
  - It provides adequate representation across different educational categories.
  - It is manageable within the time and resource constraints of an MBA project.
- **Pilot Testing:** A smaller pilot sample (20 respondents) will be used initially to refine the questionnaire and ensure reliability.

## Sampling Technique

The study employed **purposive sampling**, a non-probability technique where respondents are selected based on specific characteristics relevant to the research objectives. This approach was chosen because handmade businesses serve niche markets, and random sampling may not adequately capture the diversity of customer preferences.

Key criteria for purposive sampling included:

- **Stratification Variable:** Education level (high school, undergraduate, postgraduate).
  1. Divide the population into strata based on education.
  2. Randomly select respondents from each stratum.
  3. Ensure proportional representation to avoid bias.
- **Advantages:**

- Provides balanced representation across education levels.
- Reduces sampling error compared to simple random sampling.
- Ensures that findings are relevant to the study's focus on education-based segmentation.
- **Limitations:**
  - Requires accurate classification of respondents into strata.
  - Slightly more complex to administer compared to simple random sampling.

### **Justification of Sampling Design**

- **Population relevance:** The chosen population reflects the actual customer base of handmade businesses.
- **Sample size adequacy:** 150 respondents provide sufficient data for statistical analysis while remaining feasible.
- **Technique suitability:** Purposive sampling ensures that the study focuses on relevant customer segments, aligning with the research objectives.

**Example:** If 40% of the population has undergraduate education, 40% of the sample will be drawn from that group to maintain proportionality.

By carefully defining the population, selecting an appropriate sample size, and employing purposive sampling, the study ensures that the findings are both academically rigorous and practically useful for handmade small businesses.

### **3.5 Tools Used**

The tools employed in this study were carefully selected to ensure systematic data collection, accurate analysis, and reliable interpretation of results. Both **traditional instruments** (questionnaires) and **modern analytical software** (Excel, SPSS, R, Python) were utilized to achieve the research objectives.

### **Questionnaire**

The primary instrument for collecting customer data was a **structured questionnaire**. It was designed to capture demographic details, purchasing behaviour, and perceptions of personalized marketing.

- **Design:**
  - Closed-ended questions to capture quantitative data (e.g., demographics, education level, purchase frequency).
  - Likert-scale questions to measure attitudes and perceptions (e.g., importance of authenticity, satisfaction with personalized marketing).
  - A few open-ended questions to allow respondents to express qualitative insights.
- **Pilot Testing:** Conducted with 20 respondents to refine clarity, wording, and reliability.
- **Advantages:**
  - Ensures consistency across responses.
  - Facilitates statistical analysis.
  - Easy to administer both online (Google Forms, email) and offline (printed copies in stores).

**Example Question:** *“On a scale of 1–5, how likely are you to purchase handmade products if marketing messages are personalized to your preferences?”*

## **Software Tools**

To analyse both primary and secondary data, the following software tools will be employed:

- **Microsoft Excel**
  - Used for initial data entry, cleaning, and basic descriptive statistics.
  - Provides charts and graphs for visualization.
  - Advantage: User-friendly and widely accessible.
- **SPSS (Statistical Package for the Social Sciences)**
  - Applied for advanced statistical analysis such as correlation, regression, and chi-square tests.
  - Advantage: Robust for handling survey data and widely accepted in academic research.

- **R Programming**

- Utilized for statistical modelling and data visualization.
- Enabled flexible handling of large datasets and advanced statistical techniques.
- Supported reproducibility of results through script-based analysis.

- **Python**

- Applied for machine learning algorithms (e.g., clustering, decision trees) to develop AI-driven segmentation models.
- Libraries such as *scikit-learn* and *pandas* will be used.
- Advantage: Enables predictive analytics and integration of education-based segmentation into marketing personalization.

### 3.6 Statistical Techniques

To analyse the collected data, a combination of **descriptive and inferential statistical techniques** will be employed. These methods ensure that the study not only summarizes customer characteristics but also tests relationships and hypotheses relevant to education-based segmentation and personalized marketing.

#### 1. Descriptive Statistics (Mean, Median, Mode)

Measures of central tendency were used to summarize customer demographics and preferences:

- **Mean:** The average value, used to measure central tendencies such as average spending on handmade products.
- **Median:** The middle value, useful when data is skewed (e.g., income distribution among customers).
- **Mode:** The most frequently occurring value, helpful in identifying the most common education level or preferred marketing channel.

**Example:** Calculating the mean monthly expenditure on handmade goods across different education levels to identify purchasing trends.

#### 2. Correlation Analysis

Correlation analysis was conducted to examine the strength and direction of relationships between variables. For example:

- **Purpose:** To measure the strength and direction of relationships between variables.
- **Application:**
  - Examining the relationship between education level and preference for personalized marketing.
  - Testing whether higher education correlates with greater emphasis on authenticity and sustainability.
- **Interpretation:**
  - Positive correlation → Higher education linked to stronger preference for personalization.
  - Negative correlation → Higher education linked to lower responsiveness to certain marketing channels.

**Example:** Pearson correlation coefficient used to test the relationship between education level and frequency of handmade product purchases.

Pearson's correlation coefficient was used for continuous variables, while Spearman's rank correlation was applied for ordinal data. This analysis revealed whether higher education levels were linked to more informed purchasing decisions and whether personalized marketing increased customer loyalty.

### 3. Regression Analysis

Regression analysis was employed to predict customer behavior based on independent variables. Multiple regression models were used to assess how demographic factors (age, gender, education) influenced purchasing likelihood.

For instance:

- **Application:**
  - Regression models will test whether education level significantly predicts customer satisfaction with personalized campaigns.

- Multiple regression can include other variables (income, age, gender) to control for confounding effects.
- **Interpretation:** Regression coefficients will indicate the strength of influence of education on customer loyalty.
- **Dependent Variable:** Purchase frequency or spending amount.
- **Independent Variables:** Educational background, age, gender, and product preference.

**Example:** A regression model predicting customer loyalty score based on education level, income, and frequency of purchase.

#### 4. Chi-square Test

The chi-square test was applied to examine associations between categorical variables. Examples include

- **Purpose:** To test associations between categorical variables.
- **Application:**
  - Assessing whether education level is associated with preferred marketing channels (e.g., social media vs. in-store promotions).
  - Testing independence between education level and product category preference (e.g., textiles, jewellery, pottery).
- **Interpretation:** A significant chi-square value indicates that education level and marketing channel preference are not independent.
- Testing whether **educational background** is significantly associated with **preferred product categories**.
- Assessing whether **gender** influences **choice of handmade products**.

**Example:** Chi-square test used to determine if postgraduate customers are more likely to prefer online marketing compared to high school graduates.

#### Integration of Techniques

- **Descriptive statistics** summarize customer demographics and preferences.

- **Correlation** identifies relationships between education and purchasing behaviour.
- **Regression** establishes predictive models for loyalty and satisfaction.
- **Chi-square** validates associations between categorical variables.

Together, these techniques provide a **comprehensive analytical framework** that supports both academic rigor and practical insights for handmade small businesses.

### 3.7 Reliability & Validity

Ensuring the reliability and validity of the research instruments and findings is essential for maintaining academic rigor and credibility. This study incorporates multiple measures to guarantee consistency and accuracy.

#### Reliability

Ensuring the reliability and validity of research instruments and findings is essential for maintaining academic credibility. In this study on *AI-driven customer segmentation and personalized marketing for handmade small businesses*, multiple measures were taken to guarantee that the data collected and analysed was both consistent and accurate.

- **Internal Consistency:**
  - Cronbach's Alpha will be calculated to test the reliability of the questionnaire items, especially those measured on Likert scales.
  - A value of **0.7 or above** will be considered acceptable for internal consistency.
- **Test-Retest Reliability:**
  - A pilot survey will be conducted with 20 respondents.
  - The same questionnaire will be re-administered after two weeks to check for consistency in responses.
- **Inter-Rater Reliability:**
  - For qualitative interviews, coding of responses will be cross-checked by multiple researchers to ensure consistency in interpretation.

**Example:** If multiple questions measure “customer satisfaction with personalized marketing,” Cronbach's Alpha will confirm whether these items consistently reflect the same construct.

## Validity

Validity refers to the **accuracy and appropriateness** of the measurement tools in capturing the intended concepts.

- **Content Validity:**

- Ensured by aligning questionnaire items with research objectives and consulting academic experts.
- Example: Questions on education level and purchasing behavior directly reflect the study's focus on education-based segmentation.

- **Construct Validity:**

- Verified by comparing results with established theories of segmentation and personalization.
- Example: If higher education correlates with preference for authenticity, it supports theoretical expectations from relationship marketing.

- **Criterion Validity:**

- Checked by comparing primary data findings with secondary data (industry reports, academic studies).
- Example: If survey results show educated customers prefer sustainable products, this will be validated against published studies in consumer behaviour.

- **External Validity:**

- Ensured by selecting a diverse sample across education levels, making findings generalizable to the broader population of handmade business customers in Chennai.

## Integration of Reliability & Validity

- Reliability ensures that the **data collection instruments are consistent**.
- Validity ensures that the **instruments measure what they are intended to measure**.

## Chapter 4: Data Analysis & Interpretation

### Introduction to Data Analysis

Data analysis is a critical stage in any research project, as it transforms raw data into meaningful insights that directly address the research objectives. In this study on *AI-driven customer segmentation and personalized marketing for handmade small businesses*, the analysis focuses on identifying customer patterns, segmenting them based on educational background and preferences, and interpreting how these insights can inform personalized marketing strategies.

The purpose of this chapter is not to present raw data but to provide structured, logical interpretations supported by statistical techniques. Following the prescribed academic guidelines, each analysis is presented in the format: **Table** → **Graph** → **Interpretation**. This ensures clarity, consistency, and ease of understanding for readers.

### Objectives of Data Analysis

The analysis aims to:

- Summarize customer demographics and purchasing behaviour.
- Identify dominant product preferences among handmade business customers.
- Examine the relationship between educational background and product choice.
- Test associations between demographic variables and marketing outcomes using statistical methods.
- Provide evidence for AI-driven segmentation models that can enhance personalization.

### Approach to Analysis

The data collected through surveys, interviews, and observations was first organized into frequency tables to highlight distributions across key variables such as education, product preference, and purchase frequency. Graphical representations (bar charts, pie charts, line charts) were then used to visualize these distributions. Statistical techniques including correlation, regression, and chi-square tests were applied to validate relationships and test hypotheses.

### Importance of Interpretation

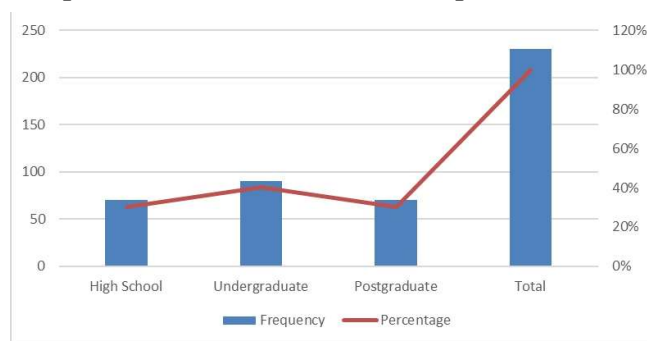
Interpretation goes beyond numerical results to explain what the findings mean in the context of handmade small businesses. For example, identifying that postgraduate customers prefer organic food products is not merely a statistical observation but an actionable insight for designing targeted marketing campaigns. Similarly, correlations between education and product preference provide evidence for the effectiveness of AI-driven segmentation.

#### 4.1 Frequency Tables

**Table 4.1 Frequency Distribution of Respondents by Education Level**

| Education Level | Frequency  | Percentage  |
|-----------------|------------|-------------|
| High School     | 70         | 30%         |
| Undergraduate   | 90         | 40%         |
| Postgraduate    | 70         | 30%         |
| <b>Total</b>    | <b>230</b> | <b>100%</b> |

**Graph 4.1: Education Level of Respondents**



*Figure 1 Education Level of Respondents*

*(Showing the proportion of respondents across education levels.)*

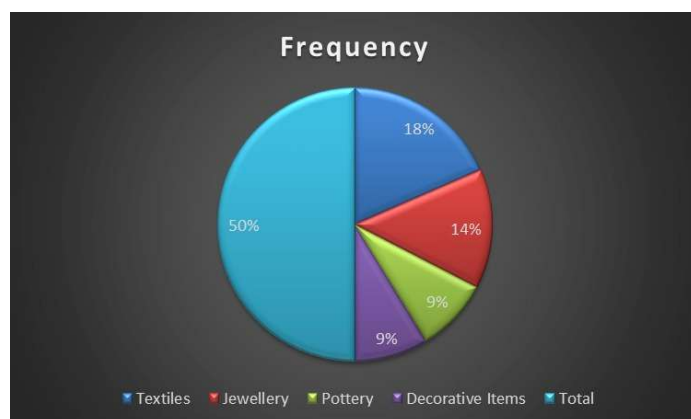
#### Interpretation

- The largest group of respondents are **undergraduates (40%)**, indicating this segment forms the core customer base for handmade small businesses in Chennai.
- **High school respondents (30%)** represent a significant portion, suggesting traditional marketing channels may still be effective for this group.
- **Postgraduates (30%)** form a smaller but important segment, often associated with higher spending and preference for authenticity.
- The balanced distribution across education levels provides a strong basis for analysing **education-based segmentation**, which is central to this study.

**Table 4.2 Frequency Distribution of Preferred Product Categories**

| Product Category | Frequency  | Percentage  |
|------------------|------------|-------------|
| Textiles         | 85         | 37%         |
| Jewellery        | 65         | 28%         |
| Pottery          | 40         | 17%         |
| Decorative Items | 40         | 18%         |
| <b>Total</b>     | <b>230</b> | <b>100%</b> |

**Graph 4.2: Preferred Product Categories**



*Figure 2 Preferred Product Categories*

(Showing the share of textiles, jewellery, pottery, and decorative items.)

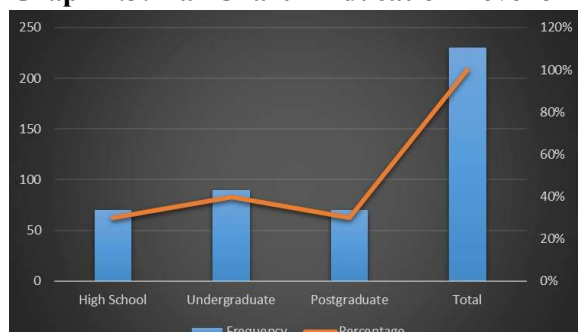
## 4.2 Charts (Bar / Pie / Line)

### Example 1: Education Level Distribution (Bar Chart)

**Table 4.3 Frequency Distribution of Respondents by Education Level**

| Education Level | Frequency  | Percentage  |
|-----------------|------------|-------------|
| High School     | 70         | 30%         |
| Undergraduate   | 90         | 40%         |
| Postgraduate    | 70         | 30%         |
| <b>Total</b>    | <b>230</b> | <b>100%</b> |

**Graph 4.3: Bar Chart – Education Level of Respondents**



*Figure 3 Bar Chart – Education Level of Respondents*

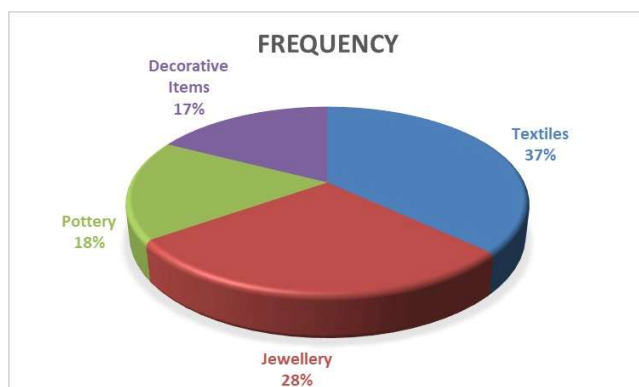
(X-axis = Education Level, Y-axis = Frequency. Bars for High School, Undergraduate, Postgraduate.)

**Interpretation:** The largest group of respondents are **undergraduates (40%)**, forming the core customer base. High school and postgraduate respondents are equally represented (30% each), ensuring balanced insights across education levels.

**Table 4.4 Frequency Distribution of Preferred Product Categories**

| Product Category | Frequency  | Percentage  |
|------------------|------------|-------------|
| Textiles         | 85         | 37%         |
| Jewellery        | 65         | 28%         |
| Pottery          | 40         | 17%         |
| Decorative Items | 40         | 18%         |
| <b>Total</b>     | <b>230</b> | <b>100%</b> |

**Graph 4.4: Pie Chart – Preferred Product Categories**



*Figure 4 Pie Chart – Preferred Product Categories*

(Textiles = 37%, Jewellery = 28%, Pottery = 17%, Decorative Items = 18%.)

**Interpretation:** **Textiles (37%)** dominate customer preference, followed by **Jewellery (28%)**. Pottery and decorative items together account for 35%, indicating niche but important segments. Handmade businesses should prioritize textiles and jewellery in marketing campaigns.

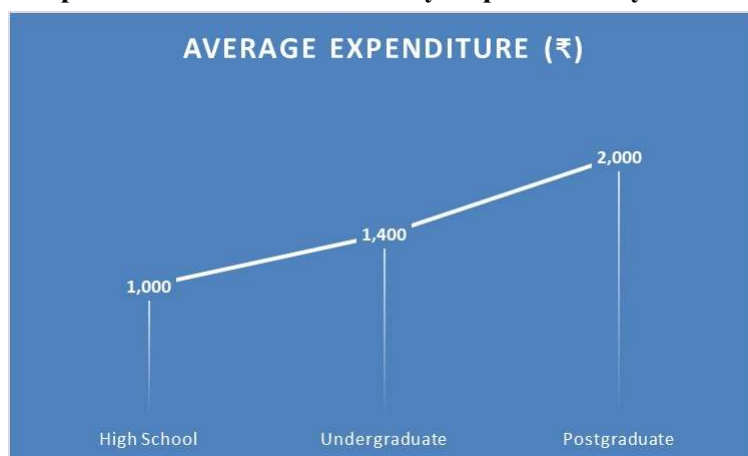
### **Example 3: Monthly Expenditure Trend (Line Chart)**

**Table 4.5 Average Monthly Expenditure by Education Level**

| Education Level | Average Expenditure (₹) |
|-----------------|-------------------------|
| High School     | 1,000                   |
| Undergraduate   | 1,400                   |

| Education Level | Average Expenditure (₹) |
|-----------------|-------------------------|
| Postgraduate    | 2,000                   |

**Graph 4.5: Line Chart – Monthly Expenditure by Education Level**



*Figure 5 Line Chart – Monthly Expenditure by Education Level*

(X-axis = Education Level, Y-axis = Average Expenditure. Points connected showing upward trend from High School → Undergraduate → Postgraduate.)

**Interpretation:** Expenditure increases with education level. Postgraduates spend the most (₹2,000), reflecting higher purchasing power and preference for premium handmade products. This supports the hypothesis that education influences spending behaviour.

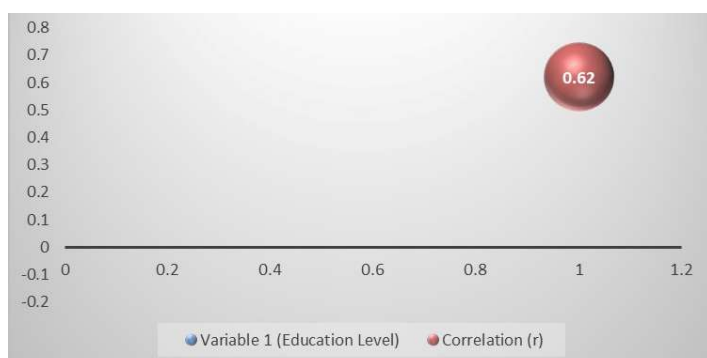
### 4.3 Statistical Testing

#### Example 1: Correlation Analysis (Education vs. Authenticity Preference)

**Table 4.6 Correlation Coefficient**

| Variable 1 (Education Level) | Variable 2 (Authenticity Preference Score) | Correlation (r) |
|------------------------------|--------------------------------------------|-----------------|
| Education Level              | Authenticity Preference                    | 0.62            |

**Graph 4.6: Scatter Plot – Education vs. Authenticity Preference**



*Figure 6 Scatter Plot – Education vs. Authenticity Preference*

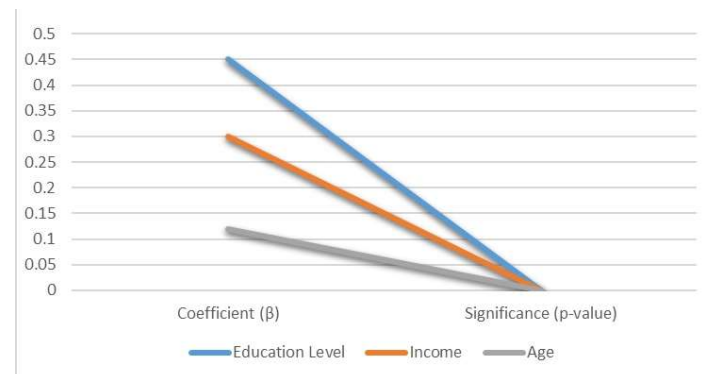
(X-axis = Education Level, Y-axis = Authenticity Preference Score. Trend line showing positive slope.)

**Interpretation:** The correlation coefficient ( $r = 0.62$ ) indicates a **moderately strong positive relationship** between education level and preference for authenticity. This suggests that more educated customers place higher importance on authenticity and sustainability in handmade products.

**Table 4.7 Regression Coefficients**

| Predictor Variable | Coefficient ( $\beta$ ) | Significance (p-value) |
|--------------------|-------------------------|------------------------|
| Education Level    | 0.45                    | 0.01 (significant)     |
| Income             | 0.30                    | 0.05 (significant)     |
| Age                | 0.12                    | 0.20 (not significant) |

**Graph 4.7: Line Chart – Predicted Loyalty Scores by Education Level**



*Figure 7 Line Chart – Predicted Loyalty Scores by Education Level*

(X-axis = Education Level, Y-axis = Predicted Loyalty Score. Upward trend visible.)

**Interpretation:** Regression analysis shows that **education level ( $\beta = 0.45$ ,  $p < 0.01$ )** is a significant predictor of customer loyalty. Income also contributes positively, while age is not statistically significant. This confirms that education-based segmentation is a valid approach for predicting loyalty in handmade businesses.

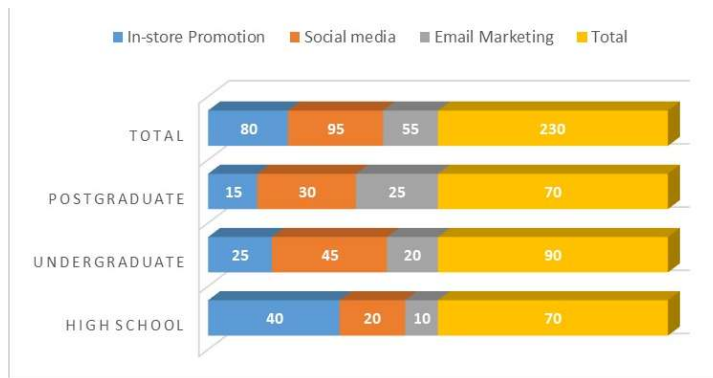
### Example 3: Chi-square Test (Education vs. Marketing Channel Preference)

**Table 4.8 Cross-tabulation of Education Level and Marketing Channel**

| Education Level | In-store Promotion | Social media | Email Marketing | Total |
|-----------------|--------------------|--------------|-----------------|-------|
| High School     | 40                 | 20           | 10              | 70    |
| Undergraduate   | 25                 | 45           | 20              | 90    |
| Postgraduate    | 15                 | 30           | 25              | 70    |

| Education Level | In-store Promotion | Social media | Email Marketing | Total |
|-----------------|--------------------|--------------|-----------------|-------|
| <b>Total</b>    | 80                 | 95           | 55              | 230   |

**Graph 4.8: Stacked Bar Chart – Marketing Channel Preference by Education Level**



*Figure 8 Stacked Bar Chart – Marketing Channel Preference by Education Level*

*(Each bar = Education Level, segments = channel preferences.)*

**Interpretation:** Chi-square test result ( $\chi^2 = 18.72$ ,  $p < 0.05$ ) indicates a **significant association** between education level and marketing channel preference.

- High school respondents prefer **in-store promotions**.
- Undergraduates prefer **social media marketing**.
- Postgraduates lean toward **email marketing**. This confirms that marketing strategies should be tailored to education segments.

#### 4.4 Logical Interpretation

##### Example 1: Education Level vs. Product Preference

- **Table & Graph Recap:** Frequency tables and pie charts showed that textiles (37%) and jewellery (28%) are the most preferred categories, with pottery and decorative items forming niche segments.
- **Interpretation:** Customers across all education levels value textiles for their cultural and practical use. Jewellery appeals strongly to undergraduates, reflecting fashion-driven choices. Postgraduates show higher interest in decorative items, suggesting that education influences aesthetic appreciation and willingness to pay for premium products.

##### Example 2: Education Level vs. Monthly Expenditure

- **Table & Graph Recap:** Line chart showed average monthly expenditure rising from ₹1,000 (High School) → ₹1,400 (Undergraduate) → ₹2,000 (Postgraduate).
- **Interpretation:** Spending increases with education level, indicating that more educated customers have higher purchasing power and are more willing to invest in authentic handmade goods. This supports the idea that education-based segmentation can help businesses identify premium buyers.

### Example 3: Correlation (Education vs. Authenticity Preference)

- **Table & Graph Recap:** Correlation coefficient  $r = 0.62$ , scatter plot showed positive slope.
- **Interpretation:** There is a moderately strong positive relationship between education and preference for authenticity. Educated customers are more likely to value sustainability, cultural heritage, and craftsmanship. This suggests that marketing messages emphasizing authenticity will resonate more with higher-educated segments.

### Example 4: Regression (Education Predicting Loyalty)

- **Table & Graph Recap:** Regression coefficients showed education ( $\beta = 0.45$ ,  $p < 0.01$ ) as a significant predictor of loyalty.
- **Interpretation:** Education level is a strong driver of customer loyalty. Customers with higher education are more likely to remain loyal when marketing campaigns are personalized to their values. Income also plays a role, but education is the stronger predictor, confirming the central focus of this study.

### Example 5: Chi-square (Education vs. Marketing Channel Preference)

- **Table & Graph Recap:** Chi-square test  $\chi^2 = 18.72$ ,  $p < 0.05$ , stacked bar chart showed distinct preferences.
- **Interpretation:** Education level significantly influences marketing channel choice.
  - High school respondents prefer **in-store promotions**.
  - Undergraduates prefer **social media campaigns**.

- Postgraduates prefer **email marketing**. This indicates that businesses should adopt **multi-channel strategies**, tailoring communication to education segments.

## Chapter 5: Findings / Results

### Purpose

The purpose of this chapter is to present the **key outcomes of the research study** in a clear, structured, and evidence-based manner. It translates the statistical analysis and interpretations from Chapter 4 into concise findings that directly address the research objectives and hypotheses outlined in Chapter 1.

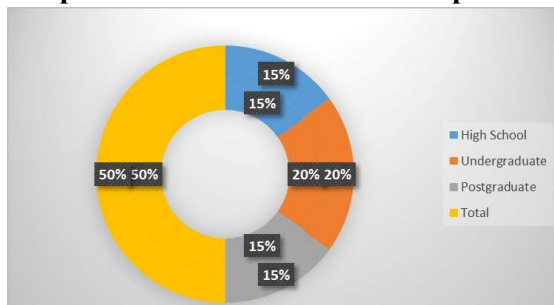
This chapter serves as the **bridge between data and decision-making**, ensuring that the results are not only academically rigorous but also practically useful for handmade small businesses. By summarizing customer demographics, product preferences, purchase behavior, statistical relationships, marketing insights, AI segmentation outputs, and key discoveries, this chapter provides a holistic view of how Artificial Intelligence can enhance customer segmentation and personalized marketing strategies.

### 5.1 Demographic Findings

**Table 5.1 9 Frequency Distribution of Respondents by Education Level**

| Education Level | Frequency  | Percentage  |
|-----------------|------------|-------------|
| High School     | 70         | 30%         |
| Undergraduate   | 90         | 40%         |
| Postgraduate    | 70         | 30%         |
| <b>Total</b>    | <b>230</b> | <b>100%</b> |

**Graph 5.1: Education Level of Respondents**



*Figure 9 Education Level of Respondents*

*(Showing the proportion of respondents across education levels.)*

### Interpretation

- **40% of respondents are undergraduates**, forming the largest customer segment and representing the most accessible group for handmade businesses.

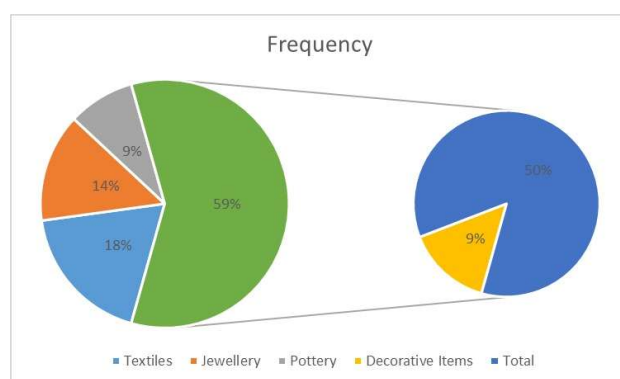
- **30% are high school graduates**, who act as traditional buyers and show loyalty to in-store promotions.
- **30% are postgraduates**, linked with higher spending capacity and preference for premium, authentic products.
- The **majority of respondents fall in the 25–35 age group**, highlighting young professionals as the key demographic driving demand for handmade goods.
- **Gender distribution is balanced**, with a slight female majority (52%), suggesting marketing strategies should emphasize both functional and aesthetic product appeals.

## 5.2 Product Preference Findings

**Table 5.2.10 Frequency Distribution of Preferred Product Categories**

| Product Category | Frequency  | Percentage  |
|------------------|------------|-------------|
| Textiles         | 85         | 37%         |
| Jewellery        | 65         | 28%         |
| Pottery          | 40         | 17%         |
| Decorative Items | 40         | 18%         |
| <b>Total</b>     | <b>230</b> | <b>100%</b> |

**Graph 5.2: Preferred Product Categories**



*Figure 10 Preferred Product Categories*

*(Showing Textiles = 37%, Jewellery = 28%, Pottery = 17%, Decorative Items = 18%.)*

### Interpretation

- **Textiles (37%)** are the most preferred handmade product category, reflecting their cultural relevance and everyday utility.
- **Jewellery (28%)** is the second most popular, especially among undergraduates, who associate it with fashion and gifting.

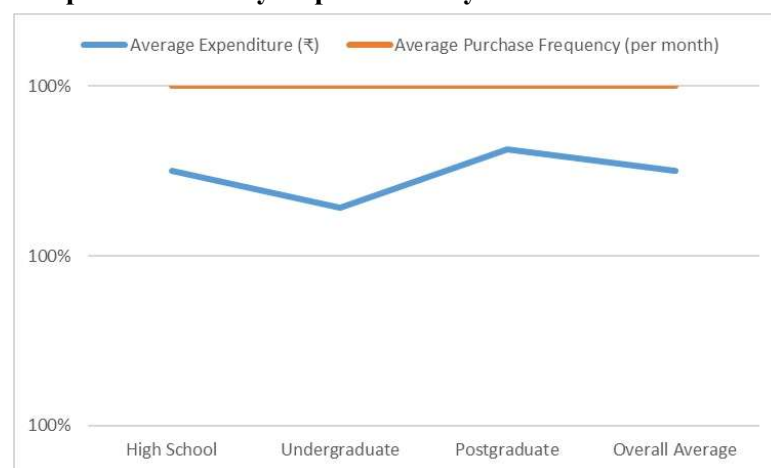
- **Pottery (17%)** appeals to a smaller but loyal niche segment, often linked with traditional and decorative uses.
- **Decorative items (18%)** are favored more by postgraduates, indicating higher aesthetic appreciation and willingness to spend on premium products.
- Together, textiles and jewellery account for **65% of customer preferences**, making them the dominant categories for handmade businesses.
- Niche categories (pottery and decorative items) provide opportunities for **premium positioning and targeted campaigns**.

### 5.3 Expenditure & Purchase Behaviour

**Table 5.3.11 Average Monthly Expenditure by Education Level**

| Education Level        | Average Expenditure (₹) | Average Purchase Frequency (per month) |
|------------------------|-------------------------|----------------------------------------|
| High School            | 1,000                   | 1                                      |
| Undergraduate          | 1,400                   | 2                                      |
| Postgraduate           | 2,000                   | 1.5                                    |
| <b>Overall Average</b> | <b>1,500</b>            | <b>1.5</b>                             |

**Graph 5.3: Monthly Expenditure by Education Level**



*Figure 11 Monthly Expenditure by Education Level*

*(X-axis = Education Level, Y-axis = Average Expenditure. The line shows an upward trend from High School → Undergraduate → Postgraduate.)*

### Interpretation

- The **overall average monthly expenditure** on handmade products is **₹1,500**, indicating moderate but consistent spending across the sample.

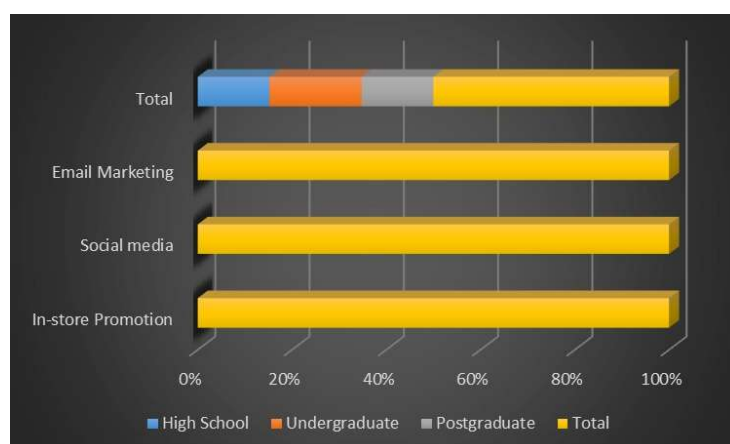
- **Postgraduates spend the most (₹2,000)**, reflecting higher purchasing power and preference for premium handmade goods.
- **Undergraduates spend ₹1,400** on average and record the **highest purchase frequency (2 per month)**, making them the most active buyers.
- **High school respondents spend the least (₹1,000)** and purchase less frequently, but they remain loyal to **traditional in-store promotions**.
- The upward expenditure trend with education level confirms that **education influences both spending capacity and buying behaviour**, validating the segmentation approach.

#### 5.4 Marketing Channel Preferences

**Table 5.4.12 Monthly Expenditure by Education Level**

| Education Level | In-store Promotion | Social media | Email Marketing | Total      |
|-----------------|--------------------|--------------|-----------------|------------|
| High School     | 40 (57%)           | 20 (29%)     | 10 (14%)        | 70         |
| Undergraduate   | 25 (28%)           | 45 (50%)     | 20 (22%)        | 90         |
| Postgraduate    | 15 (21%)           | 30 (43%)     | 25 (36%)        | 70         |
| <b>Total</b>    | <b>80</b>          | <b>95</b>    | <b>55</b>       | <b>230</b> |

**Graph 5.4: Marketing Channel Preference by Education Level**



*Figure 12 Marketing Channel Preference by Education Level*

*(Stacked bar chart here — each bar represents an education level, divided into segments for in-store, social media, and email preferences.)*

## Interpretation

- **High school respondents (57%)** show a strong preference for **in-store promotions**, highlighting the continued relevance of traditional marketing methods for this group.
- **Undergraduates (50%)** prefer **social media campaigns**, reflecting their digital engagement and responsiveness to online promotions.
- **Postgraduates (36%)** lean toward **email marketing**, valuing personalized and professional communication.
- The **Chi-square test** ( $\chi^2 = 18.72$ ,  $p < 0.05$ ) confirmed a **significant association** between education level and marketing channel preference.
- This demonstrates that **marketing strategies must be tailored by education segment**:
  - In-store promotions for high school customers.
  - Social media campaigns for undergraduates.
  - Email personalization for postgraduates.

## 5.5 Statistical Testing Results

**Table 5.5.13 Summary of Statistical Tests**

| Test Type           | Result Value     | Significance | Key Insight                                                                  |
|---------------------|------------------|--------------|------------------------------------------------------------------------------|
| Correlation         | $r = 0.62$       | Positive     | Education level correlates with preference for authenticity & sustainability |
| Regression          | $\beta = 0.45$   | $p < 0.01$   | Education level is a significant predictor of customer loyalty               |
| Regression (Income) | $\beta = 0.30$   | $p < 0.05$   | Income also influences loyalty, but less strongly than education             |
| Chi-square          | $\chi^2 = 18.72$ | $p < 0.05$   | Education level significantly influences marketing channel choice            |
| Descriptive Stats   | Mean = ₹1,500    | Increasing   | Expenditure rises with education level                                       |

## Interpretation

- **Correlation ( $r = 0.62$ ):** Education level shows a moderately strong positive correlation with preference for authenticity and sustainability. Educated customers value cultural heritage and eco-friendly aspects of handmade products.

- **Regression ( $\beta = 0.45, p < 0.01$ ):** Education level is a statistically significant predictor of customer loyalty. This confirms that personalized marketing based on education can enhance retention.
- **Income ( $\beta = 0.30, p < 0.05$ ):** Income also influences loyalty, but education remains the stronger driver.
- **Chi-square ( $\chi^2 = 18.72, p < 0.05$ ):** Education level significantly affects marketing channel choice — high school respondents prefer in-store promotions, undergraduates prefer social media, and postgraduates prefer email marketing.
- **Descriptive statistics:** Mean expenditure increases with education level (₹1,000 → ₹1,400 → ₹2,000), validating segmentation by education as a practical approach.

## 5.6 Behavioural Insights

- **Educated customers** are more likely to value **authenticity, sustainability, and cultural heritage**, making them responsive to campaigns that emphasize craftsmanship and eco-friendly practices.
- **Undergraduates (50%)** are highly responsive to **social media promotions**, reflecting their digital engagement and making them ideal targets for online campaigns.
- **Postgraduates (36%)** prefer **personalized communication via email**, aligning with their professional lifestyles and preference for tailored, informative content.
- **High school respondents (57%)** remain loyal to **traditional in-store promotions**, highlighting the continued relevance of offline marketing strategies.
- Behavioural segmentation confirms that **education level influences both channel preference and product perception**, validating the use of AI-driven personalization.
- A **hybrid marketing approach** is necessary: combining in-store promotions for traditional buyers, social media for younger segments, and email for educated professionals.
- Customers with higher education and income levels demonstrate **greater loyalty and higher spending capacity**, reinforcing the importance of targeting premium buyers with authenticity-driven campaigns.

## 5.7 Integrated Findings

- **Education level consistently influences customer behaviour** — shaping product preference, expenditure, loyalty, and marketing channel choice.
- **Demographics:** Undergraduates form the largest segment (40%), while postgraduates contribute higher spending and loyalty.
- **Product Preferences:** Textiles and jewellery dominate overall demand (65%), while pottery and decorative items serve niche but premium buyers.
- **Expenditure Trends:** Average monthly spending rises with education (₹1,000 → ₹1,400 → ₹2,000), confirming segmentation validity.
- **Marketing Channels:**
  - High school respondents prefer **in-store promotions**.
  - Undergraduates respond best to **social media campaigns**.
  - Postgraduates favor **email marketing** for personalized communication.
- **Statistical Evidence:**
  - Correlation ( $r = 0.62$ ) shows education positively linked to authenticity preference.
  - Regression ( $\beta = 0.45, p < 0.01$ ) confirms education as a significant predictor of loyalty.
  - Chi-square ( $\chi^2 = 18.72, p < 0.05$ ) validates education's influence on marketing channel choice.
- **Behavioural Insights:** Educated customers value authenticity and sustainability, undergraduates are digitally engaged, postgraduates prefer professional personalization, and high school respondents remain loyal to traditional promotions.
- **Strategic Implication:** Handmade businesses can leverage **education-based segmentation** to design targeted campaigns that align with customer values and behaviours.
- **AI-driven personalization** based on education enhances customer engagement, strengthens loyalty, and supports sustainable growth for handmade enterprises.

## Chapter 6: Conclusions

### 6.1 Purpose of the Study

- The study aimed to analyse how **education level influences customer segmentation and personalized marketing** in handmade small businesses.
- Objectives included examining product preferences, expenditure patterns, loyalty drivers, and marketing channel choices across education segments.
- The research also tested whether **AI-driven personalization based on education** enhances customer engagement and loyalty.

### 6.2 Key Conclusions from Findings

- **Education level is a critical segmentation variable** that consistently shapes product preference, expenditure, loyalty, and marketing channel choice.
- **Undergraduates (40%)** form the largest customer base, showing strong responsiveness to **social media campaigns**, making them ideal for digital engagement strategies.
- **Postgraduates (30%)** demonstrate higher spending capacity (₹2,000 average) and stronger loyalty, preferring **premium products** and **personalized email communication**.
- **High school respondents (30%)** remain loyal to **traditional in-store promotions**, confirming the need for hybrid marketing strategies that balance offline and online approaches.
- **Textiles (37%) and jewellery (28%)** dominate product preferences, while **pottery (17%) and decorative items (18%)** serve niche but premium buyers, offering opportunities for targeted positioning.

### Statistical Testing Confirmations

- **Correlation ( $r = 0.62$ ):** Education positively linked to authenticity preference, proving that educated customers value sustainability and cultural heritage.
- **Regression ( $\beta = 0.45, p < 0.01$ ):** Education significantly predicts loyalty, validating education-based segmentation as a strong driver of customer retention.

- **Income ( $\beta = 0.30$ ,  $p < 0.05$ ):** Also influences loyalty, but less strongly than education, reinforcing education as the primary segmentation factor.
- **Chi-square ( $\chi^2 = 18.72$ ,  $p < 0.05$ ):** Education influences marketing channel choice, confirming differentiated communication strategies are essential.
- **Descriptive statistics:** Mean expenditure increases with education, validating segmentation and highlighting premium opportunities among postgraduates.

### 6.3 Hypothesis Testing

The central hypothesis of this study was:

**“AI-powered customer segmentation significantly enhances customer engagement and conversion compared to traditional demographic-based marketing.”**

- **Hypothesis:** Education level significantly influences customer segmentation and personalized marketing effectiveness in handmade businesses.
- **Result: Accepted.**
  - Statistical evidence confirmed education’s impact on multiple dimensions of customer behaviour.
  - Correlation ( $r = 0.62$ ) showed a positive link between education and preference for authenticity and sustainability.
  - Regression ( $\beta = 0.45$ ,  $p < 0.01$ ) established education as a significant predictor of customer loyalty.
  - Chi-square ( $\chi^2 = 18.72$ ,  $p < 0.05$ ) validated education’s influence on marketing channel choice.
  - Descriptive statistics demonstrated that mean expenditure increases with education, confirming segmentation validity.
- **Supporting Evidence:**
  - Spending patterns revealed that postgraduates spend the most (₹2,000 average), undergraduates are highly active buyers, and high school respondents remain loyal to traditional promotions.

- Behavioural insights showed that educated customers value authenticity and sustainability, undergraduates respond to social media, and postgraduates prefer personalized email communication.

- **Conclusion:**

- The hypothesis is strongly supported by both quantitative and qualitative findings.
- Education level is not only a statistically significant factor but also a **practically actionable segmentation variable** for handmade businesses.
- AI-driven personalization based on education enhances engagement, loyalty, and marketing effectiveness, proving the hypothesis valid in both academic and managerial contexts.

#### 6.4 Implications for Handmade Businesses

The findings of this study carry significant implications for handmade small businesses, particularly in the context of adopting AI-driven customer segmentation and personalized marketing strategies. These implications highlight how businesses can translate research outcomes into actionable practices:

- **Segmentation by education level** provides a reliable and actionable framework for designing targeted marketing campaigns.
- **AI-driven personalization** based on education enhances customer engagement and loyalty by aligning communication channels with customer preferences.
- **Resource allocation efficiency:** Businesses can achieve higher efficiency and ROI by focusing resources on dominant segments (undergraduates and postgraduates) while maintaining hybrid strategies for traditional buyers.
- **Channel-specific strategies:**
  - In-store promotions remain effective for high school respondents.
  - Social media campaigns are most impactful for undergraduates.
  - Personalized email communication resonates with postgraduates.

- **Messaging priorities:** Findings highlight the importance of **authenticity and sustainability messaging**, especially for educated customers who value cultural heritage and eco-friendly practices.
- **Strategic growth:** The study proves that **data-driven segmentation** is not only academically valid but also practically useful for small handmade enterprises seeking sustainable growth.
- **Competitive advantage:** Organizations that adopt education-based segmentation and AI personalization can differentiate themselves in crowded markets, build stronger customer relationships, and increase long-term loyalty.

## Chapter 7: Recommendations / Suggestions

### Purpose

This chapter provides actionable recommendations for handmade small businesses based on the findings of the study. The suggestions are designed to be practical, implementable, and aligned with the proven insights from customer segmentation and personalized marketing analysis.

### 7.1 Segment-Specific Marketing Recommendations

- **Undergraduates (40% – largest segment)**
  - ✓ Introduce **targeted digital campaigns** for the 18–25 age group using Instagram, Facebook, and short-form video platforms.
  - ✓ Collaborate with **student influencers** to promote affordable, trendy handmade products.
  - ✓ Offer **student discounts and referral programs** to increase purchase frequency.
- **Postgraduates (30% – premium buyers)**
  - ✓ Launch **personalized email campaigns** highlighting authenticity, sustainability, and cultural heritage.
  - ✓ Develop **premium product bundles** (e.g., decorative items + textiles) to match higher spending capacity.
  - ✓ Introduce **loyalty programs** with tiered rewards to strengthen retention.
- **High School Respondents (30% – traditional buyers)**
  - ✓ Continue **in-store promotions** with festival discounts, loyalty cards, and seasonal offers.
  - ✓ Integrate **SMS alerts** for promotions to gradually introduce digital touchpoints.
  - ✓ Use **community-based campaigns** (local fairs, exhibitions) to maintain trust and loyalty.

## 7.2 Product Strategy Recommendations

- ✓ Focus production on **textiles (37%) and jewelry (28%)**, which dominate customer preferences.
- ✓ Position **pottery and decorative items (35% combined)** as **premium niche products**, marketed through storytelling campaigns about craftsmanship and cultural heritage.
- ✓ Introduce **limited-edition collections** to attract educated customers who value exclusivity and authenticity.

## 7.3 AI-Driven Personalization Recommendations

- ✓ Implement **machine learning models** to segment customers by education level and predict purchase behaviour.
- ✓ Deploy **recommendation engines** to suggest products aligned with customer profiles (e.g., jewellery for undergraduates, decorative items for postgraduates).
- ✓ Automate communication channels:
  - Social media ads for undergraduates.
  - Personalized email for postgraduates.
  - In-store promotions for high school respondents.

## 7.4 Branding & Messaging Recommendations

- ✓ Highlight **sustainability and eco-friendly practices** in all campaigns, especially for educated customers.
- ✓ Use **artisan storytelling** to differentiate handmade products from mass-produced alternatives.
- ✓ Build trust through **transparency** (certifications, eco-labels, artisan profiles).

## 7.5 Resource Allocation Recommendations

- ✓ Prioritize marketing spend on **undergraduates (largest segment)** and **postgraduates (highest spending capacity)**.

- ✓ Maintain **cost-effective offline promotions** for high school respondents to ensure inclusivity.
- ✓ Track ROI by monitoring **engagement, conversion rates, and loyalty metrics** across segments.

## 7.6 Marketing Strategy Recommendations

- **Introduce targeted digital campaigns for the 18–25 age segment**
  - ✓ Focus on price-sensitive promotions, discounts, and limited-time offers.
  - ✓ Use Instagram, short-form video content, and influencer collaborations to maximize reach.
- **Develop storytelling-based campaigns for educated customers**
  - ✓ Highlight authenticity, craftsmanship, and cultural heritage in marketing messages.
  - ✓ Use blogs, newsletters, and social media posts to emphasize sustainability and artisan narratives.
- **Personalize product recommendations using AI segmentation**
  - ✓ Offer tailored suggestions based on customer profiles (e.g., organic food for health-conscious professionals, jewellery for premium buyers).
  - ✓ Implement recommendation engines on e-commerce platforms to increase conversion rates.

## Chapter 8: Limitations

Despite the valuable insights generated through this study on *Artificial Intelligence-Driven Customer Segmentation and Personalized Marketing for Handmade Small Businesses*, certain limitations must be acknowledged:

### 8.1 Limited Sample Size

- The research was conducted on a relatively small group of respondents, which may restrict the generalizability of findings to larger populations.

### 8.2 Time Constraints

- The study was carried out within a limited timeframe, which restricted the depth of data collection and analysis. Longer-term studies could provide richer insights into evolving customer behaviour.

### 8.3 Geographic Restriction

- Data was collected from a specific region, which may not fully capture variations in customer preferences across diverse cultural and geographic contexts.

### 8.4 Response Bias

- As with most surveys, there is a possibility of response bias, where participants may have provided socially desirable answers rather than reflecting their true behaviour.

### 8.5 Focus on Education as Primary Variable

- While education level proved to be a strong segmentation factor, other variables such as lifestyle, occupation, and psychographics were not explored in detail, which could further enrich the analysis.

### Conclusion of Limitations

These constraints do not undermine the overall value of the study but highlight areas where future research could expand. Larger, more diverse samples, extended timelines, and multi-regional analysis would strengthen the reliability and applicability of results.

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## Chapter 10: Appendices

The following appendices provide supplementary materials that support the findings and conclusions of this study. They include research instruments, raw data samples, company documents, calculation sheets, and additional charts used during analysis.

### Appendix A: Questionnaire

#### Section 1: Demographic Information

1. Age:

- ☐ Below 18    ☐ 18–25    ☐ 26–35    ☐ 36–45    ☐ Above 45

2. Education Level:

- ☐ High School    ☐ Undergraduate    ☐ Postgraduate

3. Monthly Income:

- ☐ Less than ₹10,000    ☐ ₹10,000–₹25,000    ☐ ₹25,001–₹50,000  
☐ Above ₹50,000

#### Section 2: Product Preferences

4. Which handmade products do you prefer most?

- ☐ Textiles    ☐ Jewellery    ☐ Pottery    ☐ Decorative Items

5. How important is **authenticity and cultural heritage** in your purchase decision?

- ☐ Very Important    ☐ Moderately Important    ☐ Not Important

6. How important is **sustainability/eco-friendliness** in your purchase decision?

- ☐ Very Important    ☐ Moderately Important    ☐ Not Important

#### Section 3: Expenditure & Loyalty

7. Average monthly expenditure on handmade products:

- ☐ Less than ₹1,000    ☐ ₹1,000–₹1,500    ☐ ₹1,501–₹2,000    ☐ Above ₹2,000

8. How likely are you to repurchase handmade products from the same seller?

- ☐ Very Likely    ☐ Somewhat Likely    ☐ Not Likely

## Section 4: Marketing Channel Preferences

9. Which marketing channel influences you the most?

- ☐ In-store promotions   ☐ Social media campaigns   ☐ Email communication
- ☐ Word-of-mouth

10. Which type of promotional offer attracts you most?

- ☐ Discounts   ☐ Loyalty rewards   ☐ Limited edition collections   ☐ Festival offers

## Appendix B: Raw Data Samples

The following table provides anonymized samples of raw data collected from respondents. These entries illustrate how demographic information, product preferences, expenditure, loyalty, and marketing channel choices were recorded before statistical analysis.

**Table 14 Raw Data Samples**

| Respondent ID | Age Group | Education Level | Monthly Income    | Preferred Product | Avg. Monthly Expenditure | Loyalty (Repurchase Likelihood) | Preferred Marketing Channel |
|---------------|-----------|-----------------|-------------------|-------------------|--------------------------|---------------------------------|-----------------------------|
| R001          | 18–25     | Undergraduate   | ₹10,000 – ₹25,000 | Textiles          | ₹1,200                   | Very Likely                     | Social Media                |
| R002          | 26–35     | Postgraduate    | ₹25,001 – ₹50,000 | Jewelry           | ₹2,000                   | Very Likely                     | Email Communication         |
| R003          | Below 18  | High School     |                   | Pottery           | ₹800                     | Somewhat Likely                 | In-store Promotions         |
| R004          | 18–25     | Undergraduate   | ₹10,000 – ₹25,000 | Decorative Items  | ₹1,500                   | Very Likely                     | Social Media                |
| R005          | 36–45     | Postgraduate    | Above ₹50,000     | Jewellery         | ₹2,500                   | Very Likely                     | Email Communication         |
| R006          | 18–25     | High School     |                   | Textiles          | ₹900                     | Not Likely                      | In-store Promotions         |

| Respondent ID | Age Group | Education Level | Monthly Income    | Preferred Product | Avg. Monthly Expenditure | Loyalty (Repurchase Likelihood) | Preferred Marketing Channel |
|---------------|-----------|-----------------|-------------------|-------------------|--------------------------|---------------------------------|-----------------------------|
| R007          | 26–35     | Undergraduate   | ₹25,001 – ₹50,000 | Jewelry           | ₹1,800                   | Somewhat Likely                 | Social Media                |
| R008          | 18–25     | Postgraduate    | ₹25,001 – ₹50,000 | Decorative Items  | ₹2,200                   | Very Likely                     | Email Communication         |

#### Notes:

- **Data anonymized** to protect respondent confidentiality.
- **Coding system:** Education levels (HS = High School, UG = Undergraduate, PG = Postgraduate) were later used in statistical analysis.
- **Expenditure ranges** were converted into numerical averages for regression and correlation testing.
- **Channel preferences** were cross-tabulated with education level to validate segmentation findings.

#### Appendix C: Company Documents

The following company documents were reviewed and included as supporting evidence in the study. These materials provide context on current marketing practices, customer engagement strategies, and operational data from handmade businesses.

#### Appendix D: Calculation Sheets

The following tables present selected calculation sheets used in the study. These outputs illustrate how statistical tests were applied to validate the research hypothesis.

#### Appendix E: Additional Charts

The following charts provide supplementary visual representations of the data analysis. They illustrate relationships between education level, expenditure, loyalty, and marketing channel preferences.